DiCiM

Digitalised Value Management for Unlocking the potential of
Circular **M**anufacturing Systems with integrated digital solutions

History of changes		
Page/section	**Nature of change and reason**	**Publication date**
Part A		
Milestone 6	Based on the regular control and compliance check of the Grant Agreement including Annexes, there is a request to change Due Date for Milestone 6, in the Part A of the Grant Agreement 101091536 — DiCiM (p. 36) from a current due date 24 M to a new due date 42 M. This M needs to be linked to relevant deliverables in the project's WP7 – especially the D7.2, where this deliverable will compile the training materials and training plans as well as identify key stakeholders and users of the integrated digital circular solutions within the demonstrators (M42).	22.3.2024
Financial information	Budget of GOR and HGE were updated.	10.12.2024
Staff Effort	Number of PMs in the WP5 for GOR and HE were updated; Number of PMs in the WP4 and WP5 for LEX were increased.	10.12.2024
Part B		
Top of part B	List of participants has been removed, it is included in part A	14.10.2022
2.2.1	In table 17 (impact path way methodology) have been reformulated specific needs of regulators (as type of audience)	14.10.2022
3.1.1	List of work packages was removed and filled to part A (annex I)	14.10.2022
3.1.	Purchase costs items – was added additional justification – general summary for traveling and meetings	14.10.2022
3.1	There have been removed tables – 3.1a, 3.1.b, 3.1c, 3.1.e and 3.1.f, they are included in part A	14.10.2022
1.2	Stream 2, Table 6, part "Automotive spare parts demonstrator" (C-ECO Gmbh) – Challenges: added further challenges based on the C-ECO specifics	14.10.2022
1.2.	Stream 2, Table 6, part "Automotive spare parts demonstrator" (C-ECO Gmbh), KPIs – added four new challenges based on the C-ECO specifics	14.10.2022
1.2	Stream 2, Table 5: Commercial printers' demonstrator (LEXMARK), KPIs – added baseline explanation	24.10.2022
4.	Added whole section Ethics from the project's approved proposal	27.10.2022

3.1.	'Purchase costs' items (travel and subsistence, equipment, and other goods, works and services) – was added proclamation regarding the confirmation that all costs related to equipment include only the depreciation costs for the mentioned equipment.	27.10.2022
1.2 and 1.2.3 3.1.4	Table 3 and Table 9: Based on the implementation of the 2nd Amendment there is added new affiliated entity Hisense Gorenje Europe, d.o.o. (HGE) to the demonstrator's activities. Added justification of costs.	22.03.2024
3.1.4	Removed other costs of HGE	10.12.2024
2.2.2	Edited number of publications in the table 20	10.12.2024

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Terminology

Acr.	Explanation	Acr.	Explanation
CVM	Circular value model	ML	Machine learning
CMS	Circular manufacturing system	AR	Augmented reality
IoT	Internet of things	OEMs	Original Equipment Manufacturers
DLT	Distributed Ledger Technologies	PIM	Product Information Management
SMRE	Small and medium sized remanufacturing enterprises	PCB	Printed circuit board

1. Excellence

The manufacturing sector is the greatest contributor to GDP in industrialized nations (14.5% of total GDP in Europe), but also it is a major consumer of material, energy, and a significant source of waste (over 370 million tonnes of waste are generated annually in Europe).[1] This negative impact can be reduced by implementing value recovery practices. Indeed, it has been demonstrated that closing the loop of the linear value chains provides multiple gains: i) **remanufacturing typically uses 85% less energy than manufacturing** and could offset more than 800,000 tons of CO2 emissions per annum at a global scale[2], ii) a shift in reusing, remanufacturing, and recycling products could create **more than half a million jobs** in the recycling industry **across Europe**[3]. Despite these advantages, currently

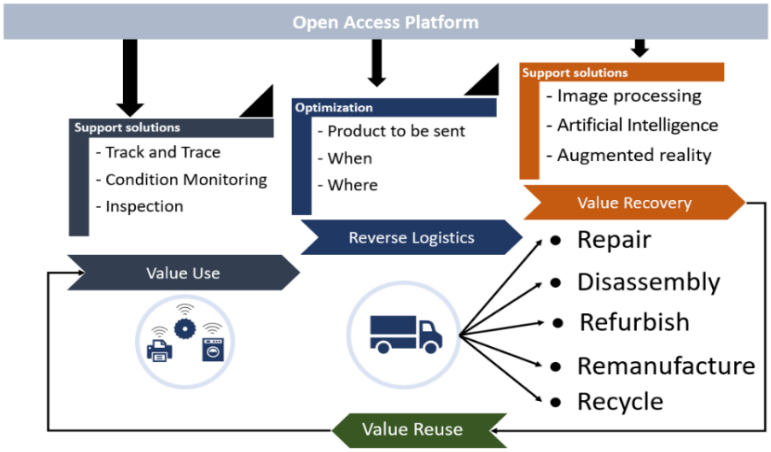

Figure 1: The schematic of the phases of the Circular Value Management systems and the integrated digital solutions that DiCiM will cover.

[1] Parker, David, et al. "Remanufacturing market study." (2015).

[2] Carbon Trust. (2014). *Innovate UK's Knowledge Transfer Network and Coventry University. Supporting Excellence in UK Remanufacturing: Stakeholder dialogue on opportunities and challenges*

[3] World Economic Forum and the Ellen MacArthur Foundation. (2014). *Towards the Circular Economy: Accelerating the scale-up across global supply chains.*

in the major industrial sectors the ratio of value recovery in the form of remanufacturing to new manufacturing is only 1.9% across Europe[1] and has the potential to become a €100 billion sector by 2030[4]. It is, therefore, quite evident that to foster the economic opportunity that is worth billions[5], the scale, efficiency, and responsiveness of the manufacturing systems for value recovery activities need to be improved significantly, with the vision to reach those of the linear manufacturing systems.

Therefore, the DiCiM project aims to develop and demonstrate integrated digital solutions that enable i) tracking, tracing, and condition monitoring of products during their use phase, ii) optimization of reverse logistics, and iii) supporting tools/solutions to enhance efficiency and responsiveness of operations during the value recovery phase.

Through these integrated digital solutions developed using state-of-the-art technologies and knowledge, the DiCiM project will address value use and value recovery, phases of **CVM**[6] (**Figure 1**) connected through an optimised reverse logistics, thus leading to a successful implementation of the **Circular Manufacturing Systems**[7] **(CMS).**

While **technology** is the main focus, the **economic and environmental performance** of the technology deployment as well as the value recovery activities will be ensured.

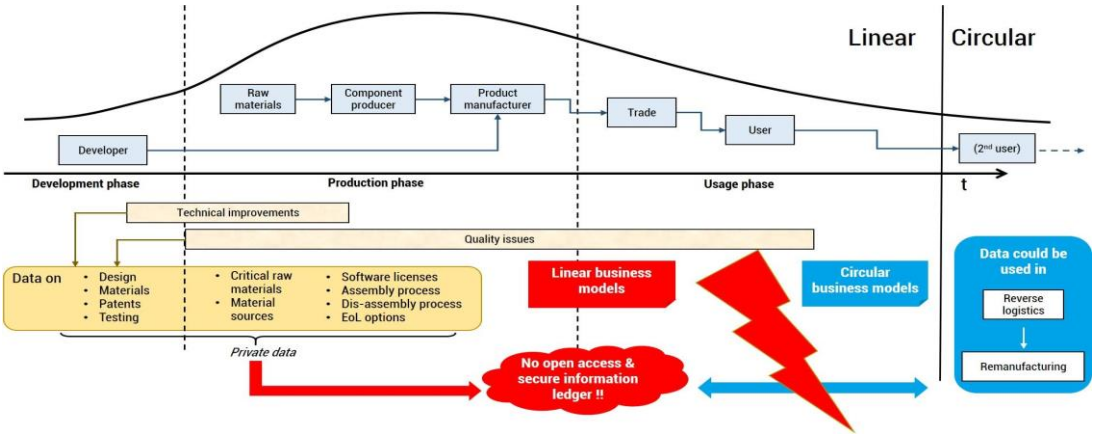

Figure 2: Traditional information management systems in the manufacturing industry are not designed for circularity and value recovery

1.1. Objectives and ambition

In the CMS, the value management approach includes phases of value creation, delivery, use, recovery, and reuse in a systemic perspective[8]. Although the value use, and value recovery phases are the main focus of the DiCiM project as the call requires, the successful implementation of CMS demands a systemic approach where **all phases of CVM are considered simultaneously and seamlessly**[9],[10]. Therefore, besides the primary focus on value use and recovery, DiCiM will maintain the necessary focus on value creation and delivery as well reuse phases to ensure the successful implementation of CMS.

DiCiM project directly addresses the expected outcomes of the call, by developing and demonstrating a set of integrated digital solutions that include an **open access digital platform** along with a range of **support solutions for managers, engineers, technicians, and operators along the value chain** to support the management and engineering activities in a Circular Economy. DiCiM is a holistic support solution that provides the necessary **infrastructure to build an agile, sustainable, efficient, and responsive production system** for recovery and reuse of products, components and materials thus reducing the dependency on imported and harmful materials.

The core goal is to facilitate and boost fundamental shifts in the way goods are produced and consumed to achieve

[4] European Remanufacturing Council. https://www.remancouncil.eu/

[5] Ellen MacArthur Foundation: *Towards the circular economy Vol. 1: Economic and business rationale for the accelerated transition*. Accessed on 07.01.16. (2012)

[6] Roci, Malvina, Niloufar Salehi, Saman Amir, Sayyed Shoaib-ul-Hasan, Farazee MA Asif, Aleš Mihelič, and Amir Rashid. "Towards Circular Manufacturing Systems implementation: A Complex Adaptive Systems perspective using modelling and simulation as a quantitative analysis tool." Sustainable Production and Consumption (2022).

[7] CMS a system that is designed intentionally for closing the loop of products/components preferably in their original form, through multiple lifecycles.

[8] Asif, Farazee. M. A., *Circular manufacturing systems: a development framework with analysis methods and tools for implementation*. Diss. KTH Royal Institute of Technology, 2017.

[9] Rashid, Amir, et al. "Resource conservative manufacturing: An essential change in business and technology paradigm for sustainable manufacturing." Journal of Cleaner production 57 (2013): 166-177.

[10] Roci, Malvina, Niloufar Salehi, Saman Amir, Sayyed Shoaib-ul-Hasan, Farazee MA Asif, Aleš Mihelič, and Amir Rashid. "Towards Circular Manufacturing Systems implementation: A Complex Adaptive Systems perspective using modelling and simulation as a quantitative analysis tool." Sustainable Production and Consumption (2022).

sustainable resource management and reduce carbon emissions, addressing the concept of the circular economy. We focus on **material and energy savings** by enabling reuse of products, components, and materials in an efficient and responsive way.

1.1.1. Objectives

It is of great importance for the DiCiM project that the development of integrated digital solutions (both the open access digital platform and the support solutions) are co-created with the users to i) allow quick feedback loops so that issues that hinder user experience can be identified and resolved; and ii) ensure rapid technology uptake in the case of companies (and beyond). Therefore, DiCiM will:

1. Develop solutions with cognitive interfaces as well as training and knowledge transfer mechanisms.
2. Develop solutions for working with the users' experience (mapping, analysing, and improvement) and for increasing engagement, especially for motivation and stimulation of pro-digital technologies implementation.

Table 1: Project Objectives

Objectives	Results
Call scope: Development of innovative concepts, methods, and tools that track and trace the status of relevant manufactured products and components, such as electronic systems and components as well as machine tools, and increase the transparency and accountability for these along their lifecycle. Where appropriate, proposals need to be able to link up with manufacturing industrial data spaces platforms, so that circular economy data can be shared with a larger set of organisations;	
Key area 1: Integrated digital solutions that combine open access digital platform for lifecycle information management and support solutions for value recovery activities	
O1: Develop open access digital platform based on relevant technologies and principles (e.g., Distributed Ledger Technologies (DLT and Gaia-X) for data infrastructure. O2: Develop IoT capability to trace and track operations and service data. O3: Develop algorithms to optimize reverse logistics. O4: Develop digital support solutions, i.e. image processing, AR and condition monitoring for decision support. O5: Implement the digital platform, algorithms and support solutions in an integrated way to support all activities of value recovery, i.e. collection, sorting, inspection, disassembly, repair/remanufacturing and reassembly.	Secure, transparent, and scalable Open access digital platform based on open-source components to increase available spare parts by 30% annually. Reverse logistics cost reduction between 30% to 50%. Increased efficiency in value recovery by 30%. Increased spare part recovery rate between 20% (washing machines) and 150% (refrigerators). OEMs, service organisations, suppliers, repair shops, used parts (core) brokers and logistics support providers, connected on one platform.
Call scope: Inclusion and handling of real-time production data in analysis software and tools, notably for decision making and control, as well as knowledge management.	
Key area 2: ML based AI for product information management and supporting decision-making for different actors of the CMS.	
O6: Apply ML for real-time decision support for optimized reverse logistics including product collection, inspection, and sorting. O7: Integrate Image processing and AR platform to visually assist with product inspection, sorting, repair and reassembly.	Necessary data is available ion real-time for personnel on the field to reduce human error related to sorting, recovery, etc, by 90%. Reduce sort and test costs at sorting centre by 25%. Increase refurbished units by 50%.
Call scope: Demonstration of the support tools in at least two different realistic production environments with a clear target of improving quality and sustainability with significant economic value.	
Key area 3: Demonstrate the impact of Integrated digital solutions in the value use and value recovery phases of the CVM.	
O8: Perform whitegoods demonstrator: washing machine (Dem1). O9: Perform whitegoods demonstrator: refrigerator (Dem2). O10: Perform consumer electronics demonstrator (Dem3) O11: Perform automotive spare parts demonstrator (Dem4). O12: Evaluate the economic and environmental performance	Testing in a real production environment, increasing good part recovery by M48: **Dem1** increased recoveries by 20,000 units /year. **Dem2** increased returns by 70,000 units /year. **Dem3** increases refurbished units by 50% /year. **Dem4** increases used spare parts recovery by

of the demos. O13: Assess the business case and upscaling potential of the demos. O14: Evaluate replication potential at the sector level.	30% /year. Further KPIs are in **Table 3** to **Table 6**. Platform and support solutions testing and validation making the developments ready for market launch.
Call scope: Research must build on existing standards or contribute to standardisation. All projects should build on or seek collaboration with existing projects.	
Key area 4: Integration and creation of standards and synergies with other relevant European, national or regional initiatives, funding programmes and platforms.	
O15: Contribute to standards such as ISO/TC 323 Circular Economy. O16: Engage with DigiPrime and CirThread projects. O17: Exploit APRA and European Remanufacturing Association's network as well as other networks (such as APPLiA and DigitalEurope, and Gaia-X association) that the partners are a member of.	Community of early adopters in the whitegoods industry and consumer electronics industry - network/ecosystem (helix) of at least 200 stakeholders by M36. Emerging standards and data format for open access platform.
Call scope: The support tools need to work with the user, and training and knowledge transfer, cognitive interfaces, as well as acceptance and uptake will be vital in the solutions proposed.	
Key area 5: Human centred approach and acceptance uptake through training, dissemination and exploitation activities	
O18: Co-creation and regular validation of support tools with end-users. O19: Training programs for support tools with end-users. O20. Develop an open network facilitating the adoption of the integrated digital solutions. O21. Disseminate demonstration results. O22. Mechanism of citizens and end-user engagement through collaboration.	4 tailored training programs ensuring knowledge transfer, uptake, and user acceptance. 2 research articles publications, motivating actors from other manufacturing sectors to use the integrated digital solutions. 5 researchers with upskilled careers circular economy and 2 new transdisciplinary/trans-sectoral collaborations.

Do not significant harm principles of the project: In line the research and innovation activities do not support, or include any activity that could make significant harm to any of the six environmental objectives identified with the Article 17 of the EU Taxonomy Regulation. Compliance is assessed both for activities carried out during the present project as well as for the expected life cycle impact of DiCiM at the commercialisation stage, to be covered with a complementary project.

DiCiM supports the achievement of the Made in Europe Partnership General Objectives. Namely: 1. Ensuring European Leadership & manufacturing excellence 2. Achieving Circular and climate-neutral manufacturing 3. Mastering the Digital transformation of the manufacturing industry 4. Creating Attractive added-value manufacturing jobs

1.1.2. Ambition

Value recovery is the process of reusing materials of a used product back into the production of a like-new product. The challenges include high cost of resources, complex parts design, limited used parts availability, lack of internationally accepted definitions and protocols, poor design of reverse logistics networks, and poor consumer perceptions.

The efficiency and responsiveness of value recovery operations depend heavily on the quality and quantity of returning products (spare parts availability) as well as

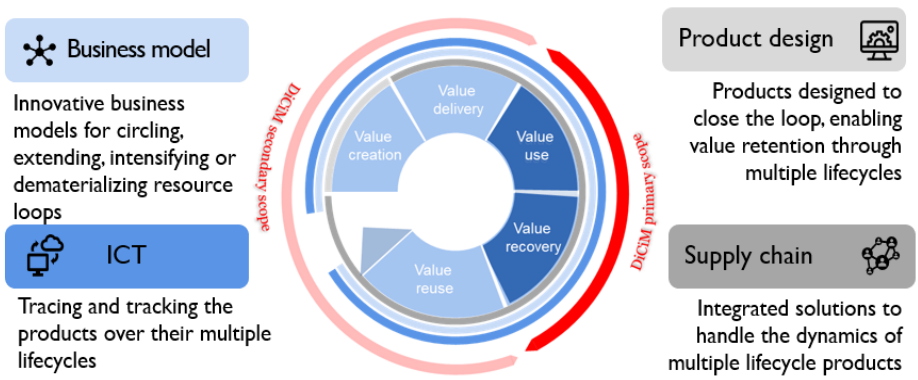

Figure 3: Value management phases in CMS and the focus of the DiCiM project.

the accessibility of support solutions to carry out activities. Knowing where used products/parts are and in what conditions can help optimize the reverse logistics process, which involves managing returns, dealing with any leases or refurbishments, and potentially buying surplus goods and materials. Tracking, tracing, and condition monitoring can provide with real time data about the location and condition of products. After the return of products, the processes of inspection, sorting, disassembly/dismantling, testing and repair/refurbish/remanufacturing/recycling start.

We consider the implementation of the circular manufacturing systems and CVM (Circular Value Model) as the cross functional integration of the four areas where their mutual dependencies and interactions are considered and understood as shown in Figure 3:

Area 1 ICT tools: Digital technologies to support CMS value phases with available technologies such as DLT principles for data collection, management and sharing among the value chain actors; IoT for tracing, tracking and condition monitoring; Big Data principles to systematically analyse and extract data collected through sensors and images; ML based AI to support decision making and Image processing and Augmented Reality (AR) to assist operations directly involved in value recovery and reuse activities

Area 2 Supply chain: Integrating both forward and reverse supply chains with the support of the ICT tools that can solve companies' and consumers' concerns regarding traceability and data protection in a tamper-proof manner. To demonstrate how the integrated (forward and reverse) supply chain can be influenced and optimised through the product design consideration below.

Area 3 Product design: This means designing the information management platform and products with condition monitoring capability with the idea of facilitating the refurbishment, remanufacturing, and possible upgrade of the product. To understand and feedback to the designer on how product and data management systems should be designed.

Area 4 Business model: To understand and demonstrate how the product design and supply chain features supported by the ICT tools can trigger the emergence of new business models. It also includes dissemination actions to enhance public acceptance and the possibility of scaling up the value reuse business.

Solutions currently used for supply chain communication and legal/ethical compliance monitoring are developed to support linear systems, none of them addresses the issues of closing the loop of products/materials, for multiple lifecycles. Furthermore, solutions do not allow inter-organization communication and sharing of data among multiple stakeholders in a value chain.

The DiCiM project will follow the ResCoM[11]-framework and explore a systemic and systematic approach, considering interdependencies among business model, product design and supply chain aspects, including ICT as a key enabler, to overcome the classic challenges of uncertainty in quality, quantity, and timing of returning products. DiCiM project will demonstrate the developments in the production environment with four case studies selected from the automotive, whitegoods and electronics sectors which represent 68% of the total manufacturing sectors in Europe (more info in section 2). The testing and validation of the developments will be carried out as shown in **Figure 4**, which is expected to create a full functioning holistic solution adaptable and applicable to other sectors.

[11] Resource Conservative Manufacturing (ResCoM) is a research initiative and also an EC funded research project (www.recoms.eu) promoting circular manufacturing systems.

1.1.3. Positioning of the project

The project builds on the results of the ReCiPSS [12] project which explored for 4 years the success factors for circular manufacturing systems, successfully implementing two cases with different OEMs showing different levels of control over their value chains: one case with full control, and one case with partial control (more info in section 1.2.3). The major technology results are outlined below in relation to their TRL at project start and completion.

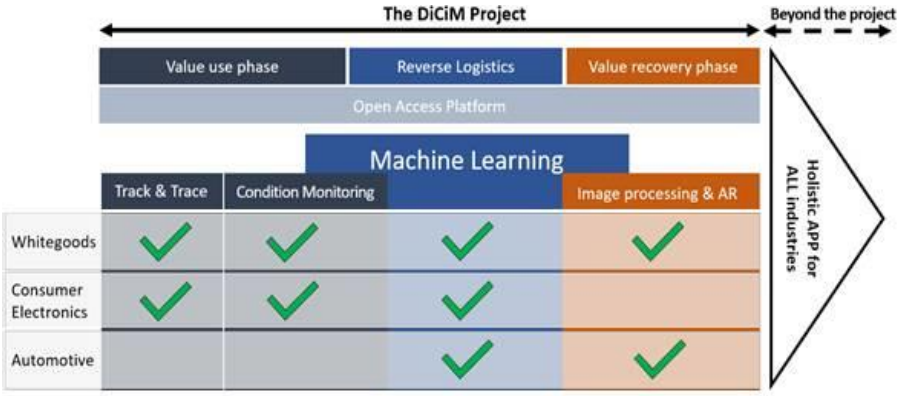

Figure 4: DiCiM plans a demonstration of the developments in the real production environments of four companies in three major industrial sectors.

Table 2: Project TRL maturity

Output	Leader	TRL start	TRL end
An open access digital platform for lifecycle information management.	SIG	4	6
Extend the capability of the ReCiPSS spare parts data management platform	C-ECO	4	6
Develop condition monitoring capability of the washing machines	GOR, HGE	4	6
Develop IoT capability for tracking and tracing of washing machines and printers with lifecycle data collection, monitoring and sharing	GOR and LEX	4	6
Apply ML for achieving longevity and optimising reverse logistics	ULFS	4	6
Apply ML based image processing for inspection, sorting, testing and value recovery operations	IRIS	4	6
Apply AR to optimize value recovery operations	TUC	4	6

1.2. Methodology

1.2.1. Overall methodology

The project is organized in 4 activity streams running in parallel during the project:

Stream 1 (Digitalised value management system) To support and interconnect Areas 1, 2 and 3 of the CMS, this stream covers the creation of the integrated digital solution with the design and development of an open access digital platform for lifecycle information management and the support solutions to sustain the implementation of circular manufacturing systems. The stream includes software and technology specification, architecture, development, and integration [WP1, WP2, WP3 and WP4], built upon the principles and research from previous projects (contributions in **Table 9**).

Stream 2 (Synergy of the demonstrators) will deploy the open access digital platform and the support solutions with existing information management systems in four cases from the three selected value chains (whitegoods, consumer electronics and automotive parts) [WP5]. The stream also includes software/hardware integration and deployment support, data collection/analysis/evaluation, business case validation and environmental impact analysis [WP6 and WP7]. To foster synergies between demonstrators, a joint taskforce will be created and will host regular meetings between the teams implementing the four demonstrators. This Stream integrates Areas 1, 2, 3 and 4 of the CMS.

Stream 3 (Integration and creation of standards) which relates to both Areas 3 and 4. This stream ensures use of relevant standards in the development, identification of gap in standards and finally propose new standards. Standards provide the fundamental building blocks for product and process development, but they also make it easier for consumers to understand and compare competing products, generating, in turn, additional business

[12] EC funded research project www.recipss.eu

models or expanding existing ones. Standards are also necessary for easier and fairer trade between geographies because the same specifications are adopted for use in different countries. It includes (but not limited to) proper labelling of remanufactured products, creating, and promoting universally developed and accepted standards on reman to raise public awareness [important contributions from WP6 and WP7. Dedicated efforts in T8.4 from WP8].

Stream 4 (Exploitation & Dissemination) Directly related to Area 4. We will compile the results of the demonstrators in high impact collaterals for broad dissemination [rest of tasks in WP8]. We will base our communication and dissemination strategy in the triple helix model, including interaction between academia (the university), industry and government, to foster economic and social impact, with both individual (Table 15) and Joint (table 16) communication actions. Specifically, partners Crowdhelix (CHX), Automotive Parts Remanufacturers Association (APRA) and the European Reman Council (ERP) to provide significant contributions. More information in sections 2.2.2 and 0.

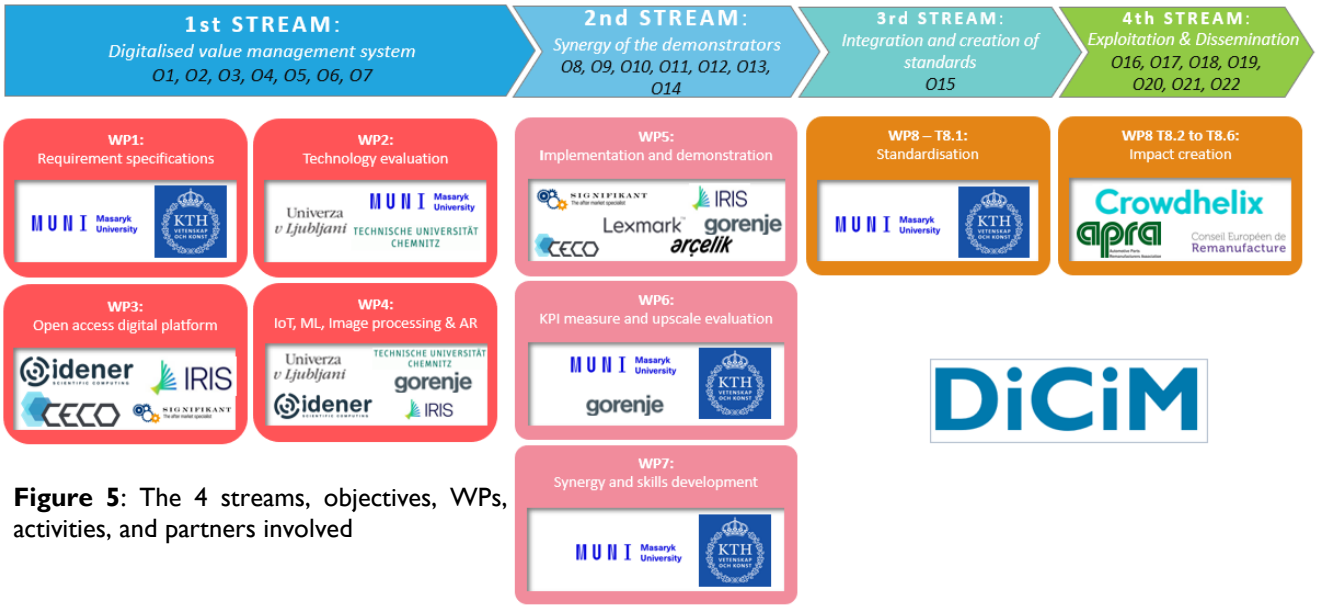

Figure 5: The 4 streams, objectives, WPs, activities, and partners involved

Stream 1: Digitalised value management system

Digitalised value management system in DiCiM means seamless integration of the support solutions with the digital platform for lifecycle information management to enhance the efficiency of value use and recovery phase together with the reverse logistics.

As electrical machines are used in different applications, prognostics play an essential role in improving system safety, reliability and enabling predictive maintenance decision making. Having an integrated health management and diagnostic strategy becomes an important part of a systems' operational life cycle. Here, data-driven prognostic usually deploys machine learning algorithms to identify the pattern and anomalies within the raw signals. **Figure 6**[13] depicts the three main types of maintenance that can be applied: corrective, preventive, and predictive maintenance. Based on the results from ReCiPSS project and identified needs from the industrial partners (see section 1.2.3), we find condition monitoring as an essential part to enable preventive and predictive maintenance and, hence, make activities like repair or refurbish more attractive and profitable, especially for small and middle-sized remanufacturing enterprises (SMREs). Condition monitoring is also an essential element to ensure the collection of the right products, in the right quality and quantity at the right time.

Generally, stresses that impact electrical machines' operation can be classified into four main categories (Thermal, Electric, Ambient, and Mechanical). The appearance of faults depends mainly on the products' parameters, such as machine type, size, rated voltage, etc. To increase the reliability of the machine, many parameters must be monitored. We will use a combination of IoT and ML technologies

The new "Right to repair" regulations compliance requires significant advances for manufacturers. It is especially relevant in 3 fields: 1) supporting the repair process, 2) ensuring the supply of spare parts and 3) monitoring of appliance part status.

[13] Kudelina, K.; Vaimann, T.; Asad, B.; Rassõlkin, A.; Kallaste, A.; Demidova, G. Trends and Challenges in Intelligent Condition Monitoring of Electrical Machines Using Machine Learning. Appl. Sci. 2021, 11, 2761. https://doi.org/10.3390/app11062761

With DiCiM integrated digital solution, designers can also produce repairing instructions, including images, videos, 3D models and a mechanical way to define repairing steps and procedures. These instructions will be accessible at three levels: 1) company repair shopfloor, 2) 3rd party certified repairmen and 3) end-users. The information will be presented to these 3 levels using a combination of AR and ML to facilitate the overall repair process. At the same time, the application that is used to support the repairing process will capture key business information (e.g., damaged parts) that will be fed to the company's internal systems. Besides, the operators will be assisted by a component defect detection system to execute a quick diagnosis (using Image Processing) of specific parts. Finally, decision makers of the company will be presented with Key Indicators about of spare part stock or overall improvements. Where data and instructions related to repair is made available to the operators through the open access platform. .

Furthermore, the upskilling capabilities of AR will also be used in the automotive sector (represented in the project by C-ECO) by ensuring that identified spare parts are correctly handled in the sorting processes. This will be achieved by checking that the right part is sorted in the right container and conveying this information visually to the worker, by utilising the capabilities of AR-glasses and connecting it with the logistic management system. This will reduce re-sorting process and delivering wrongly sorted goods to customers.

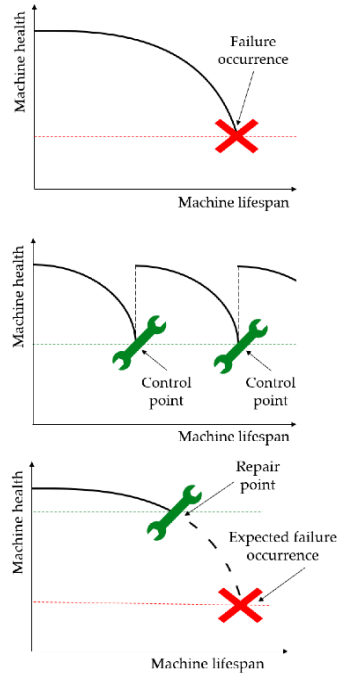

Figure 6: Maintenance types: Top: corrective maintenance Middle: preventive maintenance Bottom: predictive maintenance

Source: Kudelina et al.,2021

→ *The open access digital platform*

One of the factors preventing the standardisation and widespread adoption of open access digital platforms is their reliance on shared databases, common to the stakeholders in a value chain. This results in a poorly managed data that is easy to tamper with and the subsequent concerns regarding loss of privacy and data protection. This makes many large companies unwilling to share their data, that resort to isolated management systems, resulting in data islands along the supply chain. Distributed Ledger Technologies (DLT) has the potential to address the concern of the different actors of the value chain and allow cross-organisational collaboration.

There are many existing works in the recent literature exploring the potential of big-data and DLT technology, and how it can disrupt the supply chains[14,15]. DLT refers to the technological infrastructure and protocols that allows simultaneous access, validation, and record updating in a tamper-proof manner across a network that's spread across multiple entities or locations. At the same time, participants' privacy can be highly respected and preserved with corresponding access control measures. Transactions in a permissionless mechanism are visible to the public while the details can be restricted. The value chain actors could decide whether the stored data is public or restricted with levels of access control as well as can decide with which actors the data should be shared.

While the potential benefits are quite promising, there are some challenges hindering the development of such a technology, which we will address during the project:

Storage and speed scalability: With the number of transactions increasing day by day, the digital platforms based on DLT such as blockchain technology (just one of the potential implementations of a DLT), have become increasingly heavy as all transactions must be stored for validation. Which can be quite a time-consuming task. Currently, famous Bitcoin, for example, has raised to more than 391GB[16] of storage and can process a maximum of 7 transactions per second[17], which means that a client must wait for at least 10 minutes on average to be sure that the transaction is included. We will implement a DLT with a focus on storage and energy consumption savings.

Retrievability: Besides data storage, DLT will be used to provide data retrieval service to the stakeholders along the supply chain. However, it is non-trivial to search for data efficiently while preserving data privacy. In particular, the data stored on DLT are typically encrypted, which hinders the development of efficient search algorithms. To

[14] Benzidia, S. "E-Design: Toward a New Collaborative Exchange of Upstream E-Supply Chain." Supply Chain Forum: An International Journal 19, no. 4 (2013): 4-9.

[15] Wang, Y., M. Singgih, J. Wang, and M Rit. "Making sense of blockchain technology: How will it transform supply chains?" International Journal of Production Economics 211 (2019): 221-236.

[16] YCharts. "Bitcoin Blockchain Size," https://ycharts.com/indicators/bitcoin_blockchain_size, Accessed: 20 Feb. 2022

[17] Blockchain Council. "Top cryptocurrencies with their high transaction speeds," https://www.blockchain-council.org/cryptocurrency/top-cryptocurrencies-with-their-high-transaction-speeds/. Accessed: 21 Feb. 2022

solve this challenge, will also explore Gaia-X[18] a secure data infrastructure ecosystem developed in Europe in which data is made available, collated, and shared in a trustworthy manner. i.e., with a sovereign structure that doesn't depend on specific providers, an ownership mechanism that allows the user to always retain sovereignty over their data and using a label system. This labelling system is evidence of compliance with strict rules under the Labelling Framework. Labels could constitute facilitation in the procurement of services and an effective method to build trust between users and providers

Architecture: the main components of the open access digital platform will be a circular PIM (Product Information Management) and a circular core(used parts in the context of the automotive sector) -return-options management, see **Figure 7**. Original Equipment Manufacturers (OEMs) will share selected technical information about circular products, parts and components with the open access digital platform from their own proprietary platforms. In return they will connect their proprietary e-commerce solution to the open access digital platform and can show information from both platforms combined. The data from the PIM will also serve as the basis for the creation of core-return options to return of specific products for circular value recovery.

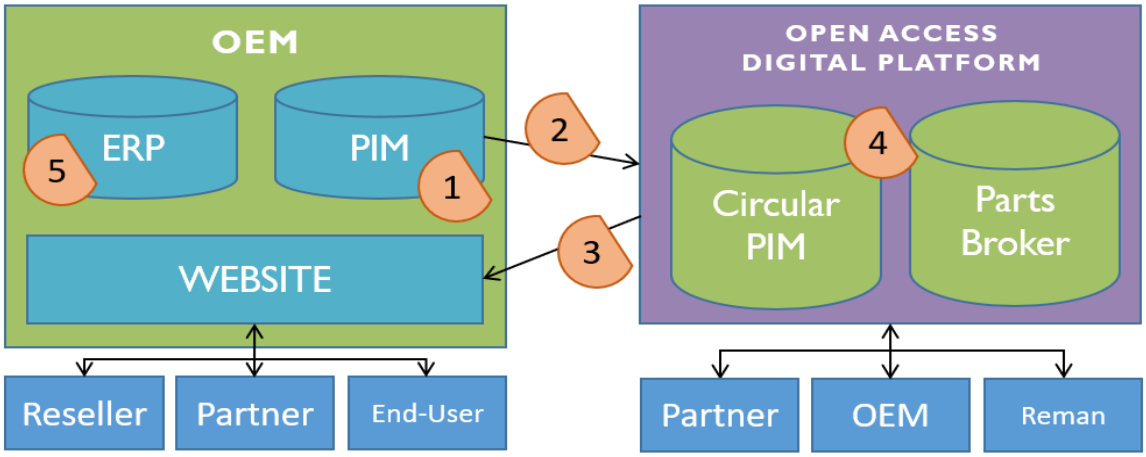

Figure 7: Preliminary architecture for open access digital platform ecosystem.

This way OEMs will be supported by the open access digital platform with the following use cases, see **Figure 7**: 1. Tag parts, components and assemblies in their own aftermarket PIM system that are eligible for remanufacturing or where remanufactured alternatives can be offered, 2. Share technical information from the OEM PIM to the circular PIM that will allow digital open access platform participants to match components with the correct products and to create matching core-return-options to fulfil the demand for used products. 3. Supply and Demand planning with respect to incentive creation for used parts return. 4. Include information from the open access digital platform in the OEM's e-commerce solution offering options for selling the old component and buying a remanufactured component. 5. Fulfilment service for used parts return to OEM/remanufacturer: pick-up of used part, identification, and quality inspection, sorting according to production needs, provide information for financial clearing.

To ensure the largest possible market for used parts, OEMs, dealers, and resellers can include circular economy options in their e-commerce solutions: 1. Besides OEM and Aftermarket parts, also remanufactured parts will be offered to the user. 2. Return options will be mentioned to return to the OEM and/or an open access participant.

→ *IoT for condition monitoring*

During the project we will leverage the power of IoT to trace, track and optimise the timing, quantity, and quality of returned products using a layered architecture, where IoT is used for sensor data processing, fusion, and analysis of all the information available:

1. The sensor data layer consists of various IoT sensors that can measure physical surroundings and capture real-time changes, measuring different parameters such as temperature, humidity, gas, Infra-Red (IR) sensors or Radiofrequency Identifiers (RFID). Taking advantage of components that may be already available in currently manufactured goods or easy to include during value recovery operations.
2. The data fusion layer is required to handle various data challenges generated from monitoring potentially heterogeneous devices. The combined data from different sources are then passed to the data analysis layer for efficient knowledge generation and decision-making.
3. The data analytic layer, in combination with AI, is what truly enables the smart functionality of DiCiM. To handle the huge volume of data generated by the IoT sensors, we will explore efficient solutions based on

[18] https://gaiax.se/

cloud computing and parallel distributed systems.

During Task 4.1 and Task 4.2 efforts will be directed towards the development of IoT infrastructure that reduces computation cost, improves network reliability, and ensures scalability.

→ *Machine Learning based artificial intelligence for reverse logistics and inspection*

Very often, field technicians must cope with situations where the problem is unknown, and this scenario forces the company they work for to send a more experienced worker to the site. A procedure of this type is, however, costly in terms of time and therefore inefficient. Likewise, the same maintenance and repair interventions represent one of the most significant critical issues for companies.

Machine Learning (ML) picks up patterns from data to arrive at predictions. Access to large data lakes, easiness of building data pipelines, increase in real-time data processing capabilities, all would make ML a powerful mainstream technology. This technology will be used in the sorting process, to assist detect whether a product can be reused, needs repair, is ready for remanufacture or has arrived at its last stage and can progress to dismantling and recycling.

Leveraging the condition monitoring capability and IoT to streamline the data (indicating the condition of the products) ML can process a large amount of data (applying big data principles) collected from the products in real-time to trigger different types of decision makings and interventions such as service intervention. This also makes possible service inspections through remote assistance. This is reflected in **reduced repair cost** and **repair time**, as the services may be planned in advance. Furthermore, a positive effect on the **reparability** and the **lifespan of the product** is expected together with an optimized reverse logistics that only bring products when the potential of value recover is the highest.

→ *Image Processing and Augmented Reality for value recovery*

We include Image Processing and Augmented Reality (AR) to compensate for the skills gap and reduce staff errors, ensure rapid and effective know-how sharing, increase the percentage of first-time resolutions and provide more accurate diagnoses. This is where this technology comes into play, helping to perform maintenance activities, virtual collaborations, and training of machinery.

The available information can also be integrated with data from IoT connected devices, which allow users to view the relevant statistics to directly evaluate the correct functioning of an asset or used in combination with the twin of an asset to be manipulated in real time and accompanied by additional information. These capabilities extend skills and allow operators to complete maintenance and service tasks more quickly and efficiently, regardless of their skill level.

Combining ML and AR in the value recovery processes enables:

1. Operational resources are more efficient thanks to clear and knowledge appropriate instructions.
2. Specialised skills and experiences are transferred to the entire organisation for shared and proactive use.
3. The training of inexperienced staff is faster and easier, thanks to collaborative learning.
4. Productivity, safety, and customer service reach new standards.

Product information is fed into the proposed framework for disassembly planning and content generation, ultimately displayed in an AR interface. Disassembly planning encapsulates information related to the disassembly sequence, and it is enhanced by the visual cues and animations provided by the content generation module. This information is ultimately presented to the operators in the AR interface.

Stream 2: Synergy of the demonstrators

DiCiM has chosen 4 cases as demonstrators: washing machines, refrigerators, printers, and automotive spare parts, representing the 3 major industrial sectors of the European Economy and waste. These demonstrators cover the whole reverse part of the circular value chain, i.e., reverse logistics, product sorting, disassembly, testing, value recovery, disassembly, and reassembly. The sectors represented by these case companies have a diverse knowledge base when it comes to value recovery activities. For instance, the automotive sector has the longest experience in value recovery activities, whereas the printing and the whitegoods industry has moderate to no experience at all respectively.

From the business point of view, the printing industry is mainly business to business (B2B) and automotive spare parts has conventionally been business to consumer (B2C), whereas whitegoods have experience from both.

B2B printing business has a tightly connected value chain with full control over the products, whereas automotive has a weakly connected value chain with no control over the products. The whitegoods sector is operating in the moderately connected value chain and with little effort (due to widespread authorised and local service business) control over the products can be easily established.

Technology-wise, the automotive spare parts sector has the most established and efficient remanufacturing infrastructure while in the whitegoods sector the value recovery capacity is limited to repair and after-sales services. In this case, the printing industry has advanced significantly but is rather far away from the automotive sector.

These diversities and differences are the areas where DiCiM will establish synergies for cross learning and knowledge sharing among the demonstrators, as part of the T7.1 (WP7) dedicated for this purpose.

Table 3: Washing machine demonstrator

Washing machine demonstrator	Gorenje and HGE
Description	As part of the ReCiPSS project, Gorenje and HGE have successfully developed new long lasting washing machines and piloted 300 washing machines in 4 European countries in a circular pay-per-wash scheme. For scaling up Gorenje plans to expand the pay-per-wash business model in different segments in the market with three distinct offers called Gold, Silver and Bronze. Gold offer includes brand new machine with premium service (i.e., service the machine instantly). Silver offer includes one-time remanufactured machines with moderate service and the Bronze offer includes two times remanufactured products with budget service (i.e. service within 1 week).
Challenge	To penetrate simultaneously in all three segments at the initial stage, Gorenje and HGE needs to have access to many used washing machines: **1)** To 'take back' (returning old machines by customers when a new machine is installed) machines and explore the 'buy back' market potential to recover spare parts for the rest of appliances. **2)** Gorenje and HGE will use its B-Stock (e.g., damaged machines during transportation and washing machines returned as warranty claim) as the source to scale up pay-per-wash business model. **3)** These collected machines will be used to source spare parts that will serve all three segments of the pay-per-wash business model and allow cost effective maintenance and refurbishment of washing machines. **4)** Due to high labour costs repair is quite expensive operation in circular pay-per-wash business operation, so it's important that every single part of it is optimized.
Solution	To optimise the return of the right product at the right time and value recovery activities to develop: 1. A smart machine capable of condition monitoring 2. IoT platform to collect information in real-time for decision making 3. AI for optimizing the timing, quantity and quality of products need to be brought back (predictive maintenance) 4. AI for decision making of what sort of value recovery activities should be undertaken both at product and component level 5. Augmented reality to support the service personnel to optimize value recovery activities
KPIs	1. Size of the B-stock WMs & Tumble dryer: 20 000 units /year 2. Increase in spare part recovery efficiency: 30% (net energy Saving (avg. 39 KWh/ WM-TD): 780 MWh) 3. Increase in spare part recovery rate: 20% (Net value Savings (avg. price of B-Sock €210): €4.2 M, Loss reduction: €190k, net Material saving (avg.146 kg/WM-TD): 2920 tons, 4. Increase in logistics efficiency 50% 5. Net CO_2 saving (avg. 2.5 tons CO_2 eq/ WM-TD): 50 000 tons CO_2 eq 6. DiCiM developments can be applied to B-sock of all other major appliances: 150 000 units/year 7. This could lead to net value savings (avg. price of B-Sock all major appliances €180): € 27 M

Table 4: Refrigerator demonstrator

Refrigerator demonstrator	Arcelik
Description	The largest refrigerator manufacturing factory of Arcelik is in Eskisehir, Turkey, which is a key player in cooling home appliances in Europe, with approximately 7M refrigerator sales globally per year. Currently, 70.000 refrigerators are collected (due to mainly return rights and warranty claims) and a major part of the refrigerators are sent directly to the recycling centre annually despite having huge potential of high quality value recovery such as refurbishment or component reuse. Due to insufficient capacity, only 10% of the collected refrigerators are inspected to enable refurbishing operations or component reuse. The insufficient capacity at Arcelik is since the decision-making mechanisms in the inspection process is currently completely manual and based on the operators' subjective assessment.

Challenge	1) Approximately 70.000 refrigerators are collected annually with capacity of inspecting only about 7000 refrigerators causing huge negative economic and environmental impacts 2) Cycle time of inspecting each refrigerator is currently **2 hours** which needs to be reduced significantly (the aim is **10-15 minutes)**. 3)The decision making of the operators are subjective and prone to making wrong decision 4) Lack of transparency and accountability of recovered components and thereby, difficult to integrate them to production and spare parts stock management
Solution	To increase the efficiency of the inspection process, support the decision-making process of the service personnel and optimize value recovery activities Arcelik will 1. Apply robot assisted image processing technique for appliance and circuit board (PCB) damage detection, and bacteria inspection 2. Apply AI for image processing to support decision making in value recovery activities
KPIs	1. Volume returns: 70 000 units (net material saving: 70000* 100 Kg = 7000 tonnes) 2. Increase in spare part recovery efficiency: 90% (10-15 minutes inspection time instead of 2 hours leading to 126 000 labour hours saved for inspecting 70 000 units) 3. Increase in spare part recovery rate: 150% (from the current state) 4. DiCiM developments can be applied to capture other major appliances; 10 million units/year 5. This could lead to net value savings: (avg, price €130): €13,6 M

Table 5: Commercial printers' demonstrator

Commercial printers' demonstrator	Lexmark
Description	Lexmark currently collects products from the markets in at least three ways. Products returning from the B2B customers at the end of the leasing contract, take back printers in connection to installing new printers and buying back a large fleet. These products have three destinies, in best cases products are refurbished and sent to new customers, harvesting spare parts and in the worst-case components are sorted for efficient recycling. These take back processes are constantly under development. .
Challenge	There is a mismatch between supply and demand causing the value recovery highly inefficient. Furthermore, there is an issue with the lack of lifecycle as well as product usage information which is essential for the decision making and planning in advance of the value recovery activity (i.e. repair, refurbish, harvest spare parts, recycle etc.) for a specific printer. Currently, Lexmark gets some data from its leased printers but for other printers (sold through wholesalers and distributors or take back printers) no data is available. Lexmark wants to know what value recovery actions are more profitable and environmentally friendly and when the printers should be taken back. In addition, Lexmark wants to achieve carbon neutrality in its operations and products (with as little compensation for CO2 as possible). Lexmark uses the LCA approach, but the LCA data for hardware and cartridge is not accessible to everyone. Currently, it is a very manual and resource-intensive process. There is an interest in making the information available to a wider audience (i.e. different roles in the company such as compliance, procurement, sales etc.) to support them in their decision making. Lexmark is interested in the whole value chain for CO2 data.
Solution	To optimise the return of the right product at the right time and value recovery activities Lexmark will develop along the project by its main entity Lexmark Hungary acting as Partner: 1. Access the lifecycle and product usage information from the built-in sensors and monitor the condition of the printers 2. IoT platform to collect information (both from the printers leased and sold through the distributors) in real-time and decision making 3. Apply AI to optimize the timing, quantity and quality of products that need to be brought back 4. Apply AI for decision making on what sort of value recovery activities should be undertaken both at the product and component level

KPIs	1. Reduce per unit reverse logistic cost:30% based on the situation from the 2021; cost information for unit reverse logistic 2. Reduce sorting as well as testing costs at sorting centre: 25%, included in fix cost in comparison to 2021. 3. Increase refurbished units: 50%, based on the comparison with the year 2021 and including the refurbished cost labor as well as refurbished material cost. 4. Decrease both average spare parts and the per unit refurbishment cost: 20%, based on the comparison with the year 2020 as baseline (cost per printer - labor, as well as cost per printer – material). 5. Increase social acceptance of refurbished and remanufactured printers/cartridges through awareness programs. Reach users via social media campaigns. 6. Increase salvaged units -dismantling, parts repair: 50%, based on the comparison with the year 2021(decrease based the increased the life-time of the printer) 7. Increase the portfolio of returned used printers:35%, based on the comparison with the year 2020.

Table 6: Automotive spare parts demonstrator

Automotive spare parts demonstrator	**C-ECO Gmbh**
Description	Remanufacturing is a widespread standard-business in automotive aftermarket, Nevertheless, there is no commercial IT-solution available in the market that exclusively supports the value recovery activities inter-organizational. Most of the large companies rely on Enterprise Resource Planning (ERP) systems which are designed for linear manufacturing systems and only include very limited functions for circular value recovery. Cases where internally developed software is used, they are designed for and limited to intra-organizational collaboration and do not support data sharing across the value chain. These "information-islands" prevent an efficient inter-organizational allocation of used parts to value preserving and recovering activities such as re-using, refurbishing or remanufacturing. As part of the ReCiPSS project, C-ECO developed an automotive part data management platform that supports core management basing on financial incentives. This platform incorporates so called core-return-options which are enabling an efficient tracking and management of financial incentives to motivate the return of used products. The concept of core-return options provides the potential to facilitate the structured return of used products for value recovery in any complex market environment where producers have only limited access to the users of their products and little control over the reverse-supply chain. Besides, C-ECO will implement AR in the sorting process to support the shop floor personnel.
Challenge	The platform currently mainly connects wholesalers from the automotive aftermarket. To fully exploit the potential of the concept of core-return-options and the ReCiPSS-platform, it is necessary to upscale the usage of the platform to increase the used spare parts return rate by a better allocation of core-return-options to used products. This can only be achieved by connecting more users and new user groups to the platform, especially functions to better involve producer/remanufacturers are needed. The key to this is to provide relevant information and efficient user interaction tailored to the business domain and informational needs of the users. This should not be limited to the commercial situation, but also provide logistic information, product-condition-documentation, and decision-support to users. It should create transparency in valuerecovery business to enable them to control the value recovery chain without being burdened with its full complexity. The core collection and sorting facilities of C-ECO collect many automotive parts from different manufacturers. After initial inspections, used spare parts are sorted based on types, manufacturers and destinations and stored in the designated boxes. Quite often the shop floor personnel places used spare parts in a wrong box, which means additional resources for corrective actions and in worst cases wrong used spare parts are delivered to the remanufacturers. C-ECO is building on the foundation of the options-platform created within the Horizon 2020 ReCiPSS-project. The intention in line with the DiCim-projects call is to provide a range of support solutions and innovative digital tools to help individuals in their business-domain in circular business processes. To achieve that we will develop the specified functions to support individuals decision, integrate them into the platform and compile and adapted them to specific targeted user-groups.

Solution	To solve this issue C-ECO needs to develop a portal compiling all relevant information for producers/remanufacturers to control the reverse value chain: 1. Realtime Reporting-Cockpit on core-return-options-situation, returned products and condition, Supply-Chain- and warehouse situation 2. Business-logic configurator for options-execution rules as decision-making support 3. Underlying configurator: machine-readable core description based on a human-centred description 4. Digital twin of core-stock: Machine learning powered supply-demand-planning and forecasting function for incentive-creation and real-time influence on sorting-decision 5. Apply AR technology where the shop floor personnel are equipped with AR glasses that prevent placing sorted used spare parts in a wrong box. This portal will contect to Signifikant's PIM-platform and the ReCIPSS options-platform to propagate used parts demand basing on core-return options towards their users.
KPIs	The open access platform can increase the offer of relevant products and used spare parts in right quality by 30% annually. For the European Automotive reman industry that would mean: 1. 1 million additional used spare parts accessible for remanufacturing in Europe (Calculated for Automotive Remanufacturing Industry). 2. Net material saving: 6000 tons (avg. used part weight 6KG) 3. Net CO_2 saving: 4500 tons CO_2-eqv (avg. CO_2 saving 4.5 kg/part) 4. € 20 million worth of value available to the value chain actors to exploit 5. Create portals for at least 2 different user-goups (e.g. "remanufacturers", "wholesalers", Buying-groups", …) 6. Validate functionalities and relevance by user-testing with users of at least 2 different business-domains per user-group (e.g. for user-group "remanufacturers": Feedback from users from logistic-department, commercial-department, sustainablility-department,…) 7. Apply AR-support in sorting-process in at least 1 of C-ECO's location and collect user-feedback 8. Reduce sorting-errors by 20%

Stream 3: Integration and creation of standards

The parts of the circular value chain corresponding to value use and recovery many challenges such as complex product design, lack of used parts availability and information, unfavourable consumer perception, absence of appropriate reverse logistic networks, and lack of standard definition. **The lack of standards is considered the most crucial barrier**[19] impacting, above all, remanufacturing.

Specifically, remanufacturing differs from other recycling procedures because it returns product quality to the OEM's performance. It builds a sustainable supply chain, beginning to end, because it requires fewer resources (40% energy, 30% material and 50% cost)[20] than new products and is environmentally friendly by design.

DiCiM proposes integrated digital solutions to support the implementation of the Circular Manufacturing Systems. However, there is a need for standards that ensure the remanufactured parts become accurate in design and quality, is consistent every time it is built, confidently inspected, and tested, and is easily maintained and retrieved due to their stored information history. Standards are also essential for the software solutions to ensure data interoperability across the value chain. Part of the efforts for DiCiM project will involve integration with current standards where they are available, identification of the gap in standards and creation of new ones, when necessary, as summarised in the following table:

Table 7: Current standards[21] and project contribution

Challenge	Standards	DiCiM project contribution
Product design	Design standards BS 8887 and BS 7000. BS8887: Design for manufacture, assembly, disassembly, and end-of-life processing (MADE) Part 3: Guide to choosing an appropriate end-of-life design strategy. ISO	AR to support repair, remanufacture, sorting etc. Design for circularity (to imagine not only how a product will be built, but how it will be rebuilt). With remanufacturing in mind,

[19] Yang, S.; Raghavendra, M.R.A.; Kaminski, J.; Pepin, H. Opportunities for industry 4.0 to support remanufacturing. Appl. Sci. 2018, 8.

[20] Chinese Characteristic Remanufacturing Industry and Its Innovation Development. Electric Welding Machine. Available online: https://en.cnki.com.cn/Article_en/CJFDTotal-DHJI201205003.htm. Accessed: 22 Feb. 2022

[21] Pratapa, P.; Subramoniam, R.; Gaur, J. Role of Standards as an Enabler in a Digital Remanufacturing Industry. Sustainability 2022, 14, 1643. https://doi.org/10.3390/su14031643

	9001:2008 or ISO 13485:2003 can evaluate the quality of recycled parts.	designers may change the materials and coatings they chose, as well as the accessibility and layout of parts.
Used parts availability	RIC ANSI standard (RIC001.1-2016[22]) identifies and inspects the device for reusability, which also includes disassembly and inspection of the parts to ensure conformance to OEM standards.	We include IoT and DLT to support product tracking that contributes to the availability of used parts.
Used parts quality management	Newly published ISO 20245:2017, cross-border trade of second-hand goods, as a point of reference for governments to establish minimum screening criteria for the trade of second-hand goods across borders.	Product tracking using IoT will provide insights into the conditions a product was exposed to including accidents, for example, before used parts acceptance.
Negative perception of remanufacturing	ISO 14020 details environmental labelling best practices: performance verification through third-party certifications and environmental labels will significantly improve the credibility of remanufacturers and convince the consumer of the recycled product quality.	DiCiM provides information to customers to improve satisfaction and create product trust in the consumer, considering differences in consumer behaviour to make more effective dissemination and communication campaigns. More info in section 1.2.5
Lack of internationally accepted definitions and protocols	RIC001.1-2016[22] addresses concerns related to a lack of a broad understanding of the remanufacturing industry and addresses each sub-process from used parts acquisition to product labelling. ISO 20245 establishes minimum screening criteria for cross-border trade of second-hand goods. Heavily affects the import of used spare parts into the different countries.	Product tracking using IoT will provide insights into the conditions of the product, cross-country operations and more to help identify and improve confidence during the international exchange of goods.
Reverse logistics network	ISO/TC 323 committee has triggered frameworks for circular economy models.	DiCiM provides a solution for OEMs, SMREs, and dealers to have their circular value chain managed as a service without losing control or being burdened with the effort and complexity of maintaining their own reverse logistics structure.
Data integration with other departments	The ISO 15926 is a standard for data integration, sharing, exchange, and hand-over between computer systems. Wide availability of data models and reference data libraries to create models.	The needs of different actors of the supply chain to be integrated into the mainstream planning using integrated ERP systems and processes.
Data infrastructure	Manufacturing Data Space initiative to support data sharing among manufacturing companies and with (service) providers, which serves as a model to show how sharing industrial data improves company operations, enhances business opportunities, and supports the transition towards a circular economy.	The open access platform will be developed as per the reference architecture model for data sovereignty, decentral data management, data economy and data governance. This will ensure that the circular economy data can be shared with organisations beyond the DiCiM project.
Skill training, talent, and technology	ISO 9001 specifies requirements for a quality management system within an organization that can deliver products that consistently meet customer and regulatory requirements and improve customer satisfaction.	DiCiM includes AR capabilities for value recovery and sorting processes

[22] RIC001.1-2016; Reman Standard. Remanufacturing Industries Council: West Henrietta, NY, USA, 2016. Available online: https://remanstandard.us/. Access on 22 February 2022

1.2.2. Use case

All the challenges of the demonstrators described in **Table 3-Table 6** have in common the need to access transparent lifecycle information and to share that information along the value chain. This also includes integration with other IT systems of the value chain actors to ensure data interoperability necessary for high quality value recovery. This common challenge (within the demonstrators and beyond) is addressed by the open access digital platform that DiCiM will develop. We highlight some key use case examples of the open access digital platform:

Table 8: Use case examples for DiCiM open access digital platform

	Digital infrastructure needs	DiCiM Development
Gorenje, Lexmark, Arcelik	OEMs must coordinate with hundreds of SMEs to receive information about the parts they supply (material compositions, sources, etc.) to enhance value recovery activities. Most of them do not have structured data repositories and supply unstructured data by telephone or e-mail, which OEMs needs to manually process, analyse, and archive. There is a high risk of incomplete or lost information.	OEMs send a secure DiCiM data request via a URL, which contains a structured template for the data collection required from its suppliers. Suppliers must enter all data in mandatory fields until they can share the (now structured) data with OEMs. The data gets immediately integrated into OEMs' internal repositories for easy storage and retrieval.
	OEMs want to store all documents and instructions related to value recovery activities in a secure platform that is connected (for data exchange) to the ERP systems but independent to avoid external parties accessing OEMs' internal platform.	OEMs store relevant information about the value recovery activities in the DiCiM platform.
Lexmark	Different divisions of Lexmark have done Life Cycle Assessment (LCA) of products and their value chains. But the information is scattered and often difficult as well as time-consuming to access.	Lexmark makes LCA results available on the DiCiM platform and new LCA results are uploaded on DiCiM platform.
	Lexmark does not have access to the lifecycle information of the products that are sold through distributors. Although Lexmark's distributors collect and store this information on their local server, it cannot be shared due to interoperability issues between distributors' and Lexmarks' IT systems.	DiCiM creates a gateway for the distributors to share the information instantly.
Gorenje, Lexmar	To optimise the supply chain the OEMs want to keep a track of products (based on conditions and AI) that need to be brought back over time	DiCiM platform collects this information from the IoT platform and makes this amiable for the OMEs to coordinate and optimise return logistics.
Gorenje, Lexmark, Arcelik	To optimise the spare parts inventory OEMs want to keep a track of spare parts sourced from the returned products	DiCiM platform collects this information from the shop floor after components are recovered and based on this information the OEMs can create a long-term spare parts planning.
	To match the supply and demand of the refurbished and remanufactured products OEMs needs to keep a track of the inventory of the refurbished and remanufactured products.	DiCiM platform collects this information from the shop floor after products are refurbished and remanufactured and based on this information the OEMs can create a long-term spare parts planning.
Repair Shop	An independent repair shop has received a washing machine produced by Gorenje and a fridge produced by Arcelik. To efficiently repair these products the mechanic needs repair manuals.	The repair shop opens an account in the DiCiM platform and requests to get access to Gorenje and Arcelik's information about the value recovery activities. Each company review the request and prepare a digital contract for the Repair Shop. After signing the contract, the Repair Shop receives access to the desired information.

Core brokers	C-ECO offers the service to manage used parts return for producers/remanufacturers from the market basing on financial incentives. To give them control on the reverse value chain, but do not burden them with the full complexity C-ECO needs t a comprehensive point-of-information to provide supply-chain-, commercial- and product-oriented information and from where missing core-supply can be identified and propagated to the market.	C-ECO uses the planning information originating from its data out of the incentive system in DiCiM platform to identify core-demand and create core return options to cover this demand. This is done by using the Signifikants PIM-data via the DiCiM platform. The core return-options are then interlinked into Signifikants spare-parts-platform and transferred to its users in the event of relevant core-returns. Now the 'options' are available to a wider network of potential core suppliers and reman-producers can control the brokers. An individual core supplier in Signifikants spare parts platform gets directly connected to this demand in DiCiM platform where the 'options' are available during his ordering process.
DiCiM platform owner	Signifikant sees that there is a shift in the business model of the software industry. Selling a license and later on selling maintenance and integration services is becoming history for the software industry.	To capture the current trend Signifikant makes the DiCiM platform an open-access platform that is free for users to open an account as well as assess publicly available information. The users can also store relevant information for free. Signifikant charges the user of the platform a fee only when information is exchanged with other users on the DiCiM platform. Platform affordable to SMEs business model.
Consumer	There is legislation in the discussion which insists that a consumer who wants to repair a product he/she should have the opportunity to do so. The OMEs need to make repair manuals and video instructions etc. available for the consumers.	The consumers open an account on DiCiM platform to access the publicly available information about the repair of appliances.

1.2.3. National and international research & innovation activities linked to the project

We have identified the most relevant projects with synergies to the DiCiM project and described how their results could feed into our project:

Table 9: National and international research & innovation activities linked to the project

Project	Partner	Project description and cross-fertilisation opportunities
ReCiPSS [H2020] 2018 - 2022	MU KTH C-ECO SIG GOR HGE	In ReCiPSS, it was explored success factors for circular manufacturing systems in two cases where OEMs have different levels of control over their value chains: one case with full control, and one case with partial control. The project gathers 13 partners from 8 countries. It involves the study of multiple circular business models, it's virtual modelling, simulation, and demonstration with two industry-driven large-scale key industries: white goods and automotive spare parts.
DigiPrime [H2020] 2020 - 2023	C-ECO	DigiPrime has the mission to develop a new concept of Circular Economy digital platform overcoming current information asymmetry among value-chain stakeholders, in order to unlock new circular business models based on the data-enhanced recovery and re-use of functions and materials from high value-added post-use products with a cross-sectorial approach. In that context, C-ECO has developed a service for logistic optimization and reverse-network configuration together with some partners out of the consortium. This includes a mobile app which shall be used in the demonstration-phase by automotive workshops to check in used parts combined with information on the last usage environment of the part (the vehicle). Data is shared with remanufacturers via the platform and use the data for logistic pick-up-planning and network-optimization. C-ECO is the WP leader for the demonstration WP. The project involves 20 use cases covering different European industrial sectors (automotive, renewable energy, electronics, textile, construction).
CircThread [H2020] 2021-2025	GOR HGE	The CircThread project's main objective is to unlock access to data now in silo's and enhancing it as decision information for actors across and outside the extended product life cycle. The core is to create data linkages between product chain, value chain, asset

		chain and life cycle chains based on a Product information Catalogue and enable information exchanges via data contracts governed by secure and reliable management standards. All the developments create synergies with DiCiM project.
SerBiT [Eureka] 2016-2019	KTH, GOR, SIG	Project to develop and verify an IT-platform to support the manufacturing industry in implementing Service-Based Business Models (SBBM) in the context of Circular Manufacturing Systems (CMS). The goal of the project has been achieved through three unique developments, an IT platform, a prototype of a smart washing machine and an IoT platform. All these developments work simultaneously in collecting, processing, and sharing data in real-time and thereby, enable the implementation of SBBM.
ResCOM [FP7] 2013-2017	KTH, GOR	Innovative methodology and software platform for the industrial implementation of closed-loop manufacturing systems. The generic requirements of the platform proposed for implementation of closed-loop manufacturing and the data model used for the management of the lifecycle of multiple products platform will be adapted in the DiCiM project
5G-TIMBER [HORIZON EUROPE]	CHX	5G-TIMBER aims to conduct advanced field trials of the most representative and innovative data-driven material, production, and installation flow by examples of small-volume manufacturing, across 4 industries under the umbrella of the wood industry. Crowdhelix will create a new helix during the 5G-TIMBER project with links and synergies with the DiCiM Digital helix.
EIBA (Funding reference 033R226)	C-ECO	C-ECO is the co-ordinator of this project which has received funding from German federal ministry of education and research. Its goal is to use digital sensor-data, machine-vision, and artificial intelligence to identify used automotive parts for remanufacturing. The system will be demonstrated in C-ECO's network of inspection centres.
C-SERVEES [H2020] 2018 - 2022	ARCE & LEX	Lexmark is part of the project consortium, contributes to the project by setting demo related to a circular economy business model, agreed within the C-Servees project. The project aims to boost the circular economy, boosting remanufactured and refurbished products in the electrical and electronic sector. Various activities have been carried out to examine new possibilities (e.g.: dismantling demo, refurbished printer testing with customers, customer surveys, recycled material testing)

1.2.4. Interdisciplinarity and integration of social science and humanities

To create effective change that spurs a sustainable and responsive production environment and supply chains, we built a consortium with knowledge from different disciplines. They combine expertise from different aspects of the value-chain, including:

Table 10: Capabilities of the partners

Domain \ Partner	MU	KTH	TUC	ULFS	IDE	IRIS	SIG	C-ECO	GOR	ARCE	LEX	CHX	APRA	ERC
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Data processing & analysis			X	X	X	X		X	X					
Machine learning				X	X	X	X	X						
Distributed Ledger Technology				X	X		X							
Reverse Logistics	X							X	X	X	X			
Societal acceptance	X								X		X	X	X	X
Circular economy and supply chain	X	X		X			X	X	X	X	X	X	X	X
Environmental assessment		X									X			
Industrial production		X	X						X	X	X			
Re-manufacturing & recycling		X								X	X			
Business processes	X	X			X	X	X	X	X			X	X	X

1.2.5. Gender dimension

Consumers, through their actions and purchasing habits, can have considerable influence on how products are developed, delivered, used, and potentially reused. In this context, there are multiple studies that highlight a strong gender dimension regarding consumption patterns and environmental concerns. For instance, when selecting electronic products, women prefer those that have an end-of-life feature (namely reuse, remanufacture, or

recycle)[23]. In general, women tend to follow a more sustainable approach in their attitudes, being more responsive to the ecological and environmental considerations of product purchased and consumed, including their reusability when there is a choice[24,25,26]. The OECD points out how "women can be key drivers of change, as consumers and as decision-makers, in both the public and private sectors. Integrating gender equality is essential for the successful implementation of a circular economy strategy"[27]

An attractive digital tool that has the capacity to support the engineering of a Circular Economy must support and promote sustainable consumption practices that prioritise waste minimisation by extending product life. This can only be achieved with a clear understanding of consumer behaviour. **Targeting gender roles and behavioural preferences** in consumption as well as waste generation and prevention could be a key pillar in the **transition to the circular economy not only by reducing waste but also by addressing some gender inequalities**[28].

We regard it as crucial to address the differences in consumption patterns at the end-user level, as well as different attitudes, perceptions, environmental attitudes. The project brings importance through joint and specific dissemination actions. To engage all societal actors in the creation of good circular systems, through awareness-raising on sustainable consumption and encouraging participation in leadership and managerial roles. Data and information collection tasks (such as co-creation workshops) will ensure that opinions and inputs of all genders are collected and given equal weights. Both WP 8 and 9 will ensure that the gender dimension is ensured in project management and exploitation and dissemination of results.

1.2.6. Open Science practices

The Open Science approach, following findable, accessible, interoperable, and reusable (FAIR) principles, has been considered as the main pillar when planning the measures to maximise the reach of communication and dissemination activities for DiCiM project. In the context of this project, scientific information refers to article pre-prints, peer-reviewed articles in scholarly journals, articles in conference proceedings, monographs, patents, and research data (data underlying publications, curated data and/or raw data).

CHX, the dissemination leader, will establish the dissemination strategy from the start (dissemination plan available by M6) of the project. It will include early and open sharing of research (through pre-registration, pre-prints and registered reports), research output management including data management (as described in the section below) and measures to ensure reproducibility or research outputs.

Pre-print versions of all articles will be deposited into multidisciplinary open access repositories, (mainly Zenodo, more information in **Table 11**). Once published in a scholarly journal or a conference proceeding, the article's final version will be immediately posted into the Institutional open repository. This repository will automatically feed OpenAIRE with articles to ensure the largest possible impact among researchers, policymakers and businesses. DiCiM consortium members will publish peer-reviewed scientific articles in gold open access journals listed by the Directory of Open Access Journals (DOAJ). Key partners have reserved a budget for article processing charges of gold open access publications.

Each partner will ensure open access to the deposited publication (via the repository) within six months of publication, assuming an electronic version is available without extra charge via the publisher. The partners will also ensure access to the bibliographic metadata that identify the deposited publication (including terms under the grant agreement, the name of the action, acronym and grant number; the publication date, and length of embargo period if applicable, and a persistent identifier). All scientific peer-reviewed articles will be published under Creative Commons CC BY 4.0 license. Thus, the authors retain all the copyright of the article. However, the partners will retain their copyright and grant adequate licences to publishers based on Creative Commons licenses.

1.2.7. Data management and management of other research outputs

Data management are part of WP9, which will start with the formal release of a Data Management Pan (D9.3) at

[23] Atlason, R., D. Giacalone and K. Parajuly (2017), "Product design in the circular economy: Users' perception of end-of-life scenarios for electrical and electronic appliances", *Journal of Cleaner Production*, Vol. 168, pp. 1059-1069, http://dx.doi.org/10.1016/j.jclepro.2017.09.082.

[24] Kaenzig, J., S. Heinzle and R. Wüstenhagen (2013), *Whatever the customer wants, the customer gets? Exploring the gap between consumer preferences and default electricity products in Germany*, Energy Policy, Vol. 53, http://dx.doi.org/10.1016/j.enpol.2012.10.061.

[25] Khan, N. and P. Trivedi (2015), *Gender Differences and Sustainable Consumption Behavior*, http://www.eajoutrnals.org.

[26] Bulut, Z., F. Kökalan Çımrin and O. Doğan (2017), *Gender, generation and sustainable consumption: Exploring the behaviour of consumers from Izmir, Turkey*, International Journal of Consumer Studies, Vol. 41/6, pp. 597-604, http://dx.doi.org/10.1111/ijcs.12371.

[27] OECD (2021), *Gender and the Environment: Building Evidence and Policies to Achieve the SDGs*, OECD Publishing, Paris, https://doi.org/10.1787/3d32ca39-en.

[28] OECD (2020), *Session 5, Gender-specific consumption patterns, behavioral insights, and circular economy*, 2020 Global Forum on Environment, Paris, 5-6 March, http://www.oecd.org/env/GFE-Gender-Issues-Note-Session-5.pdf.

M6. The project will follow the open data management guidelines of the Horizon Europe Programme, namely:

• The project will deposit the research data in a research data repository that facilitates linking publications and underlying data through persistent identifiers and data citations, such as EPOS data and service repository, of which the CSC IDA repository is an integral part.

• The project will enable third parties to access, mine, exploit, reproduce and disseminate (free of charge for any user) these research data by attaching Creative Commons Licences to the data deposited.

• The project will provide information via the research repository about the tools available to the users that are needed to validate the results (e.g., specialised software or software code, algorithms, and analysis protocols). Wherever possible, the project will also provide these instruments.

To maximise dissemination impact, we will include a policy to make all project deliverables labelled as "public" (**Chyba! Nenalezen zdroj odkazů.**) and related datasets freely available on the project website. A publishable v ersion of the document will be produced and shared on the project website in all other cases.

Table 11: Data management methodology and tools

DATA SUMMARY	Data purpose: Our project will collect and generate product and socio-economic data related to the Circular Economy within the target of interest in the automotive, electronics and whitegoods sectors. Relevant data will be placed in open repositories after a period of grace for the consortium members to properly curate and assess whether some results are sensitive in nature, given the strategic importance of the production environments and supply chains.
	Data origin: The project data may originate from open databases; 3rd party suppliers or industrial pilots. We will ensure that we sign formal open data sharing agreements with all parties for those datasets we would like to place in open repositories.
	Data utility: We expect that our open datasets will be useful to 1) researchers (to progress the state-of-the-art further), 2) industrial partners (to improve the data exchange among value chain actors), and 3) public authorities (to support them in defining their role as promoters, facilitators and enablers).
FINDABILITY	Discoverability: Open datasets will be made findable with persistent links and a clear metadata schema
	Persistent identifiers: We will use the ORCID reference (Open Researcher and Contributor ID) for researchers in our team. ORCIDs are valuable because they connect all researcher's outputs regardless of the author's name or affiliation changes. For digital research materials, we will use the Direct Object Identifier (DOI). It can be assigned to articles, datasets, or any other shared project output.
	Naming conventions: For public project documents, the parent file will be the title of the work package, and each work package file will contain a minutes file as well as files for each task within the work package. The minute's file will contain the meeting minutes for the work package, while the task files will contain documents pertaining to that task's activities and may potentially contain an additional minute file containing meeting minutes specific to that task.
ACCESSIBILITY	Data repository: **Zenodo**[29] **repository** has been chosen as the main platform to upload all public data sets and scientific publications. Additionally, we will consider other platforms, such as the European Open Science Cloud (EOSC), the European Data Infrastructure (EDI) and the Open Research Europe publishing platform.
	Open data and access methods: Zenodo enables FAIR practices by creating a persistent digital object identifier (DOI) for each item, making them easily citable, it complies with DataCite's Metadata Schema, which is one of the broadest cross-domain standards available and, as the servers are hosted at CERN, the data and metadata will be retained and maintained by and for the lifetime of the laboratory.
INTEROPERABILITY	Metadata vocabularies, standards and methodologies according to the purpose: (i) Descriptive metadata to help users discover and identify a resource (e.g., Title, Abstract, Author, and Keywords). (ii)Administrative metadata to manage an information resource (e.g., provenance, processing, and rights information). (iii) Structural metadata describe the relationship between parts of an information resource, especially digital resources, which often consist of multiple files. (iv) Technical metadata to document technical attributes of digital objects or resources. (v) Preservation metadata to support and document the long-term digital preservation of an information resource or digital object.
RE-USE	Documentation: All our documentation will include basic information about the data that allow for its correct interpretation and reuse by ourselves in the future and by other researchers (e.g., README file, README tab, Data Dictionary, Codebook, Commented Code, Lab Notebooks, as applicable).
	Licenses: Each dataset used in the project will be accompanied by a data license that details the permissions associated with the use of that dataset. The licenses for each dataset will be stored in their respective

[29] General-purpose open-access repository developed under the European OpenAIRE program and operated by CERN. Accessible at https://zenodo.org/

	repository, where they will be found in the respective dataset's README and metadata file.
SECURITY	Data can be made available within the project team and shared with partners and stakeholders via Zenodo. The data is stored in Zenodo's entire technical infrastructure is located on CERN's premises which is subject to CERN's legal status, which includes immunity from the jurisdiction of the national courts to ensure their independence from the individual Member States. The servers are managed according to the CERN Security Baseline for Servers, meaning e.g. remote access to our servers are restricted to Zenodo staff with appropriate training, and the operating system and installed applications are kept updated with the latest security patches.
ETHICS	GDPR compliance: The project (and its data management tools) will be fully compliant with the General Data Protection Regulation (GDPR) regulations laid out in Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons regarding the processing of personal data and the free movement of such data and repealing Directive 95/46/EC (20) and respects regulations on intellectual property rights (IPR) (24). Database access will be entirely anonymous. If a user wishes to use our open repositories, they will be required to register an account. The user must provide their email address and a password and grant their consent that their email address is processed and used for account-related communications and management. In addition, data on the type of user (i.e., industry, research, etc.) will be logged to provide the Consortium with key statistics on data access by final users. The user is free to revoke their consent at any time, which would lead to the deletion of their account.

2. Impact

2.1. Project's pathways towards impact

2.1.1. The big picture.

A circular business model is sustainable only if the value can be economically and environmentally recovered from the product.

In Europe, cross-sectoral activities to facilitate knowledge transfer and promote the industry do not exist in remanufacturing - unlike in the recycling industry. Our major competitors, the USA and China, already have a common vision and strategy for remanufacturing within their industrial landscapes[30]. European remanufacturing could lose competitiveness against these more organised economies. As a result, there is a real need for a European-level solution to encourage remanufacturing throughout Europe.

By keeping components and their embodied material in use for longer, large environmental benefits can be realised; for example, the remanufacturing of automotive components yields some 88% materials savings compared to using a new product, with an associated 53% decrease in CO_2 emitted and 56% lower energy requirement.

Further, securing reliable and undistorted access to certain 'critical' raw materials is a growing concern within the EU to reduce the high risk associated with their supply. **Remanufacturing has the ability to keep advanced materials, including those termed as critical raw materials, within the economy**.

In addition to its economic and environmental benefits, remanufacturing provides opportunities for the creation of highly skilled jobs and economic growth. Despite these positives, remanufacturing is an undervalued part of the industrial landscape and an under-recognised, sustainable industry.

Remanufacturing faces many challenges such as complex product design, high cost of resources, lack of standard definition, unfavourable consumer perception, absence of appropriate reverse logistics network and financials, and lack of spare parts availability and information[31]. These challenges prevent this process from significantly impacting sustainability and are addressed with a series of standards and frameworks. These standards make reman feasible for organizations and even advance their green agendas under the UN.

Digital technologies, including the IoT and ML, can help smooth the implementation of remanufacture standards and allow for low-cost operations, price-sensitive markets, and cheap labour costs[32]. However, standards to resolve these challenges are neglected today in research and practice. The lack of remanufacturing standards is considered the most crucial barrier today[33].

[30] Parker, D., Riley, K., Robinson, S., Symington, H., Tewson, J., Jansson, K., Ramkumar, S. and Peck, D., 2015. Remanufacturing Market Study. [online] Remanufacturing.eu. Available at: <https://www.remanufacturing.eu/assets/pdfs/remanufacturing-market-study.pdf> [Accessed 8 March 2022].

[31] Chakraborty, K.; Mondal, S.; Mukherjee, K. Critical analysis of enablers and barriers in extension of useful life of automotive products through remanufacturing. J. Clean. Prod. 2019, 227, 1117–1135.

[32] Ghoreishi, M.; Happonen, A. Key enablers for deploying artificial intelligence for circular economy embracing sustainable product design: Three case studies. AIP Conf. Proc. 2020, 2233, 050008.

[33] Pratapa, P.; Subramoniam, R.; Gaur, J. Role of Standards as an Enabler in a Digital Remanufacturing Industry. Sustainability 2022, 14, 1643. https://doi.org/10.3390/su14031643

The skills and knowledge of operators involved in value recovery operations, can also be improved by using AR technology for a process traditionally considered as a non-value-added, such as disassembly or sorting operations. This is a cost and time-efficient optimisation to improve the efficiency of the disassembly and sorting processes performed by human operators and optimize the spare parts and core-retrieval process.

There is a lack of universally accepted specifications and standards for data, products, and business processes, regarding the design, quality, manufacturing, and production of used products[30], which can lead to unauthorized collection and illegal use of scrap parts, hurting remanufacturing reputation and efforts. As a result, there are regulatory barriers preventing core parts to cross international borders due to a lack of clear distinction between remanufactured products and refurbished products, pre-owned products in existing government policies and regulations. The European Remanufacturing Network (ERN) further verifies that "An absence of an accepted legal definition was found to exacerbate this issue on a global scale"[30], presenting a prominent hindrance in advancing the value recovery industry in many European countries. It is pivotal that standards are developed and propagated internationally to provide consistency and transparency in the economy and facilitate the free movement of used spare parts (known as cores in the automotive sector) to allow remanufactured products to be more accessible.

The solutions to be developed must contribute to proper identification of remanufactured products, encouraging public procurement of equipment remanufactured to specified quality standards, creating, and promoting universally developed and accepted standards on reman to raise public awareness.

2.1.2. Addressing expected outcomes with specific project impacts.

DiCiM projects understands these challenges and addresses de expected outcomes (call topic) and support the overall EC ambition (call destination) of industrial leadership and increased autonomy in key strategic value chains.

Table 12: Expected outcomes and project impacts

Expected Outcome	Pathway Towards Impact	Impact KPI by 2030
Call topic: Provide a range of support solutions and innovative digital tools for engineers, technicians**,** and operators on the factory floor, in order to build agile, sustainable and responsive production environment and supply chains, with specific focus on areas such as material saving, repair, refurbishing, re-manufacturing, recycling, and reuse of products and components	The project includes: (1) support during the value use phase with IoT and ML to monitor the performance of core products and create effective and optimized reverse logistics schedules. (2) Support during the value recovery phase with ML and AR for initial inspection and decision assistance for engineers, technicians, and operators on the factory floor. Although the present project focuses on value reuse and value recovery, all areas of the CVM are impacted.	As explained in **Table 3** to **Table 6** by applying the integrated digital solutions the demonstrators can improve spare parts recovery efficiency and rate as well as logistics efficiency by 30%-90%, 20%-150% and 30%-50% respectively.
Call topic: Reduction of the dependency from imported raw materials or harmful materials for the European manufacturing sector (e.g. by material consumption reduction, material substitution and use of secondary raw materials);	The remanufacturing industry helps contain and circulate critical raw materials. The project contributes (1) with standard creation to facilitate the free movement of used spare parts and allow remanufactured products to be more accessible throughout the European Union. (2) With IoT to monitor the quality and quantity of core components as well as the accessibility of support solutions to carry out the activities.	EU´s domestic supply for critical raw materials will be enhanced sustainably. The four demonstrators combined have the potential of net material saving more than 16000 tons pf materials per year.
Call topic: Define specifications and standards for data, products, and/or business processes, that can be agreed and commonly used by many industrial actors and across different industry sectors; and facilitate industry agreements on circularity and sustainability through increased data exchange among value chain actors and enable the development of new types of businesses;	We envision standards covering all areas: (1) Product design, (2) Core availability, (3) Core quality management, (4) Negative perception of remanufacturing, (5) Lack of internationally accepted definitions and protocols, (6) Reverse logistics network, (7) Data integration with other departments, and (8) Skill training, talent, and technology	The project implements the existing standards (Table 7) and will contribute to creating new standards with at least the 4 demonstrators studied in the project. Social acceptance is gained through the awareness programmes reaching 5,000,000 citizens.

Call topic: Reduce the skills and knowledge gap for the actors involved.	Product information is fed into the proposed framework for disassembly planning and content generation. Disassembly planning encapsulates information related to the disassembly sequence, and it is enhanced by the visual cues and animations provided by the content generation module. This information is ultimately presented to the end-user in the AR interface.	The project will develop the knowledge base in 3 manufacturing sectors. The skill and knowledge gap among the operators that will be involved in the demonstrators and continue using the development will be reduced. Furthermore, the project will develop (WP 7) high-quality training materials for widespread application for enhancing the skill of the European labor force to improve their employability (Task 7.4)
Call destination (wider impact): Global leadership in clean and climate-neutral industrial value chains, circular economy and climate-neutral digital systems and infrastructures (networks, data centers), through innovative production and manufacturing processes and their digitization, new business models, sustainable-by-design advanced materials and technologies enabling the switch to decarbonization in all major emitting industrial sectors, including green digital technologies.	A circular business model is sustainable only if value can be economically recovered from the product. Remanufacture requires fewer resources (40% energy, 30% material, and 50% cost) compared to the production of a new product [34]. Further, remanufacturing keeps raw materials within the economy. It also provides opportunities to create high skill jobs and economic growth; and yet, despite all these benefits, remanufacturing remains an undervalued and under-recognized part of the industrial landscape. The project aims to enhance a circular economy through the digitization of the remanufacture activities, which are proven to reduce waste and avoid significant energy use and emission to air and water[35].	Our objective is to create the gold standard for value recovery in the use cases considered. The solutions developed in the project can have high replicability potential in other industrial sectors. Thus, placing European software, at the centre of the global circular economy value chain. Savings due to value recovery actions of reducing 40% energy, 30% material and 50% cost for the European manufacture. which is aligned to the estimated energy, material and cost reduction indicated in Table 3 to Table 6 for the demonstrators.

2.1.3. Scale and significance of the project's contribution to the expected outcomes and impacts.

The impacts of the project can be much higher on a longer term, once project outcomes are transferred, replicated and exploited across the EU. DiCiM integrated digital solutions fall into the Global Industry 4.0 Market projected to grow from €59 bn in 2021 to €150 bn by 2026 at a CAGR of 20.6%,[36] driven by the rising adoption of digital technologies in the manufacturing process.

Table 13: Business case

Application	Market	User Needs	Competitive Advantage
Digitalised value management system with IoT, ML and DLT support	The AI in manufacturing market is blooming - expected to grow from €1bn in 2020 to 15.2 bn by 2026 at a CAGR of 57.2% over the period (Source: Marketsandmarkets)	Companies need to make value recovery and reuse activities profitable. To achieve optimization of reverse logistics as well as efficiency and responsiveness in value recovery activities.	Digitalised value management system connected to product information management systems (including ERP and PIM) to support decision making
Value Recovery with AR (general case)	AR is revolutionizing the manufacturing industry, expected to hit €2.6 bn by 2023.	Factories want to use AR solutions to become more customer-centric, efficient, and	The first AR solution fully integrated with existing tools including a web-based

[34] Chinese Characteristic Remanufacturing Industry and Its Innovation Development. Electric Welding Machine. Available online: https://en.cnki.com.cn/Article_en/CJFDTotal-DHJI201205003.htm (accessed on 03 March 2022).

[35] Parker, D., Riley, K., Robinson, S., Symington, H., Tewson, J., Jansson, K., Ramkumar, S. and Peck, D., 2015. Remanufacturing Market Study. [online] Remanufacturing.eu. Available at: <https://www.remanufacturing.eu/assets/pdfs/remanufacturing-market-study.pdf> [Accessed 8 March 2022].

[36] Industry 4.0 Market. Source: marketsandmarkets

	A recent survey research reveals the use of AR in product and service development could deliver a US$360 billion GDP boost by 2030. (Source: Analytics insight)	responsive by troubleshooting remotely in real-time with experts, securing safety and quality and reducing expenses.	application for disassembly and value recovery and a standalone AR solution for sorting parts
Value recovery (Remanufacture)	Remanufacture differs from other recycling procedures because it returns product quality to the original equipment manufacturer's performance. the activity generates around €30bn in turnover and employs around 190,000 people[35]. The four key regions, estimated to account for some 70 % of remanufacturing value in Europe, are Germany, the UK & Ireland, France and Italy[35].	Remanufacture faces many challenges such as complex product design, high cost of resources, lack of standard definition, unfavourable consumer perception, absence of appropriate reverse logistics network and financials, and lack of core availability and information[37]. These challenges prevent this process from significantly impacting sustainability and are addressed with a series of standards and frameworks.	DiCiM is endorsed by important associations of major associations with the capacity to influence different groups: from regulatory bodies to the general public. In additions to partner CHX, advisors ERC and APRA Europe APPLiA, in collaboration with DigitalEurope and DATEurope to generate a change in consumer perception and entice stakeholder collaboration.
Conclusion: The 4 innovation streams of the project are combined to develop an industrial platform supporting value recovery with higher productivity and lower costs. The platform architecture is modular to allow for different applications (i.e., combining a subset of these capabilities to produce a more cost-effective solution for a specific domain). This also will increase the impact of the project, as individual capabilities can be packaged as independent results and exploited separately.			

2.1.4. Identifying potential barriers and addressing them with project activities.

Currently, the consortium has not identified any regulatory or compliance issues that could affect directly the project executions. However, as the value recovery of products is impacted by current regulations, standards, and the public acceptance, we have isolated three main barriers to the adoption of DiCiM:

Table 14: Potential barriers to impact and how we plan to address them in the project.

Barrier	How can it limit the project impact?	How will we address it?
Technical obstacles	DLT based solutions may not scale well on cases with large data sets or may require strong technical expertise to manage the data and its consistency.	We have committed part of the efforts during the development of the project to evaluate the state-of-the-art of DLT and explore alternatives such as Gaia-X.
Standards	As explained in **Table 7**, there is multitude of standards affecting different parts of the value recovery chain. To create significant change, it may be necessary to contribute to several standards at the same time increasing the complexity and limiting the project's impact	Important stakeholders with the ability to impact standards are part of the project (namely CHX, STR and ERC). We have also dedicated resources to this end. We will start by influencing those more accessible and critical standards and continue beyond the project.
Impact obstacles	Stakeholders are unwilling to be part of an open access platform, as it may sound threatening to some due to misconceptions of what open access means	In communication and dissemination, it will be made clear that open access does not mean open data.

2.1.5. Exploitation strategy.

→ ***Scale up plan to market***

The results of DiCiM project continue to mature after the end of this project. To this aim, a scale up plan has been designed for each exploitable technology to reach the market (TRL7-TRL9). The Open access digital platform will be developed by SIG partner, with support from IDE, IRIS, and C-ECO in WP3, and integrated with developments in WP4 done by research partners for demonstration activities in WP5. Potential future exploitation by interested partners is presented in **Table 16**. We have dedicated resources in T8.5 to produce a detailed business plan with

[37] Pratapa, P.; Subramoniam, R.; Gaur, J. Role of Standards as an Enabler in a Digital Remanufacturing Industry. Sustainability 2022, 14, 1643. https://doi.org/10.3390/su14031643

future steps as the project nears TRL6.

Table 15: Technology road to TRL9 and estimated time in years after the project.

Technology		Scale-up activities (TRL7-TRL9)	Duration
Open access digital platform		- Create API for 3rd party stakeholders to integrate with the platform - Make the platform accessible for the end-users to facilitate product returns. - Integrate training materials in the platform - Final UX/UI design that is user friendly	1.5 Years
Support solutions	ML solutions	- Scale-up of IoT infrastructure - Final training of the algorithms to refine the condition monitoring and image processing	2 Years
	AR App	- Image processing and AR applicability to further industries.	

→ ***Exploitation plan***

The consortium has identified a preliminary list of exploitable results and defined preliminary exploitation strategies for them. During the project, the consortium members will develop a business plan supporting the market introduction and replication of the project results.

Table 16: Exploitable result, interested partner and business model.

Partner	Exploitable result	Value proposition and patenting areas	Business model
SIG	Open access digital platform	The platform is available as a stand-alone solution and also to integrate into the existing IT systems to manage product lifecycle information for value recovery activities	Licencing/ subscription
LEX	ML algorithms for condition monitoring	Condition monitoring capability is available for distributors and wholesalers to offer a better service to their customers	Sell as service
IRIS	ML algorithms for image processing	Image processing capability is available for other sectors	Product and service
TUC, IDE	AR applications	DiCiM project will bring forward AR applications that can be commercialized in the automotive and white goods sectors	Product and service
C-ECO	Software solution	Right used spare parts, in the right quality, at the right time and in the right location	Used spare parts
	Software solution	A software module that can be integrated to other IT systems.	Licencing and pay-per-use
GOR LEX ARCE	Efficient product and part recovery mechanism	Low-cost products and spare parts are available long after the guarantee period is over	Offer recovered parts as spare parts
	Technology implemented	Successful implementation of technology with an indication of clear improvement in value recovery activities.	Knowledge sharing to scale up the technology internally
IDE	Appliances repairing data management	The results achieved within the development of the Gorenje use case on repairing tracking and support will be integrated in the AiCORE solution which IDE is already developing for commercial exploitation.	Software Licensing
ULFS	Condition monitoring solution	Open-source solution for condition monitoring of white goods and printers, which is applicable by other actors	MIT licence (Free)
KTH, MU, ULFS, TUC	Contents and materials for education	DiCiM project will bring forward beyond state-of-the-art knowledge and cases of technology implementation	Consultancy, training and education and research projects
KTH	Models and KPIs for performance evaluations	DiCiM will define and develop KPIs as well as models (mathematical and simulation) for evaluating socio-economic and environmental performance evaluation	

2.2. Measures to maximise impact – Dissemination, exploitation, and communication.

2.2.1. Impact pathway methodologies

Table 17: Impact pathway methodology.

Audience	Specific needs	Measures to maximise the impact

Researchers	Promote the utilisation of standards for research scope in the academic sector to ensure that in the future, these standards will be implemented in the market.	Educate and employ existing standards for manufacturing through 4 workshops and public awareness events in at least the 4 universities in the project.
Industry players	A digital solution that supports new business models based on value recovery activities, such as remanufacturing or refurbishing existing products.	200 members of Digital Helix by end of the project. Increased public awareness of value recovered products becomes positive.
Industry workers	Interaction with the tools is different depending on the worker's position. i.e., managers or operators. Different actors will need to assimilate the tool for it to be effective	Specific training, seminars, workshops and e-learning materials for executives, middle managers, engineers, and operators to reach 2,000,000 blue- and white-collar workers.
Regulators	Legal issues and frameworks that are relevant for the technologies in the context of the European Union and national regulations where the demonstrators will be operated.	Increased policy-maker awareness about the importance of CMS due to supply chain, sustainability, responsibility and economic risks linked to outside production.
Public society	Understanding novel technologies and learning about the positive impact of advanced materials	3,000,000 citizens exposed to the e-learning module about value recovery actions (including courses, social media, podcasts, YouTube, press releases, publications, national science popularisation, etc.)

All the described measures to maximise impact are covered

2.2.2. Dissemination & communication activities

For the dissemination and communication actions the project counts on the support of 3 important platforms:

- Crowdhelix (CHX): Open Innovation platform that creates links between researchers and companies. Who contributes through the creation of a Virtual community and with a detailed clustering plan (details below)
- Automotive Parts Remanufacturers Association, Europe (APRA): a global association that has been representing the interests of automotive remanufacturing companies since 1941
- The European Reman Council (ERC): represents small and large businesses from all remanufactured product sectors.

While CHX will be the main responsible for dissemination activities, APRA and ERC will be vital counselling partners to reach key stakeholders.

→ ***DiCiM Virtual Community/Ecosystem "Digital" Helix: An Impact Accelerator***

DiCiM project will strongly contribute to the already existing vibrant, international, virtual community/ecosystem, named the Digital Helix. It will be an engine for impact, exploitation, clustering, and dissemination activities.

DiCiM consortium will benefit from the **CHX network that gathers 500+ member organisations from 51 countries** (all EU member states). It has the potential to reach 8,000+ research and innovation collaborators from its existing membership base. **This network**, supported by a customised technology platform, language processing and machine learning tools, **has 40 virtual thematic areas**/clusters (called "helixes"), that cover all the research and innovation fields. Through the helix, the consortium will create a network of at least 150 new organisations (M36), including: DiCiM partners, members from existing relevant Helixes (i.e., Manufacturing, Materials, Raw Minerals, Circular Plastics helixes, etc.), researchers (i.e., experts in digital/electronic systems, circular economy, manufacturing, recycling etc. The UK "Digital Catapult" is already a member), industry actors (i.e., "IBM" is already a member, digital tools providers, key actors in the European manufacturing sector, etc.), ethic and policy makers, civil society (end users), participants of other relevant projects to promote and develop clustering activities with sister projects from previous European projects and initiatives, (i.e. projects in **Table 9**), and relevant actors/associations across Europe/International (i.e., The European Factories of the Future Research Association (EFFRA), Processes4Planet, EIT Manufacturing, EIT Digital, Smart Specialisation Platform for Industrial Modernisation (S3P-Industry), etc.). We will integrate policymakers and public administrations to provide feedback on policy measures. To ensure sustainability, CHX will commit to maintaining the Helix for at least 2 years after the project has concluded. When the project ends, the Helix will continue to be hosted on Crowdhelix's platform,

and everyone signed up during the project will be able to retain their accounts for free (as external organisations), as well as have the option for their organisations to join as full members on the network. The CHX Helix Manager will identify a member, from the Helix, that would be interested in strategically leading this Helix after the end of the project. The new helix will be supported by the CHX team. The activities performed in the Helix are shown in **Figure 8**

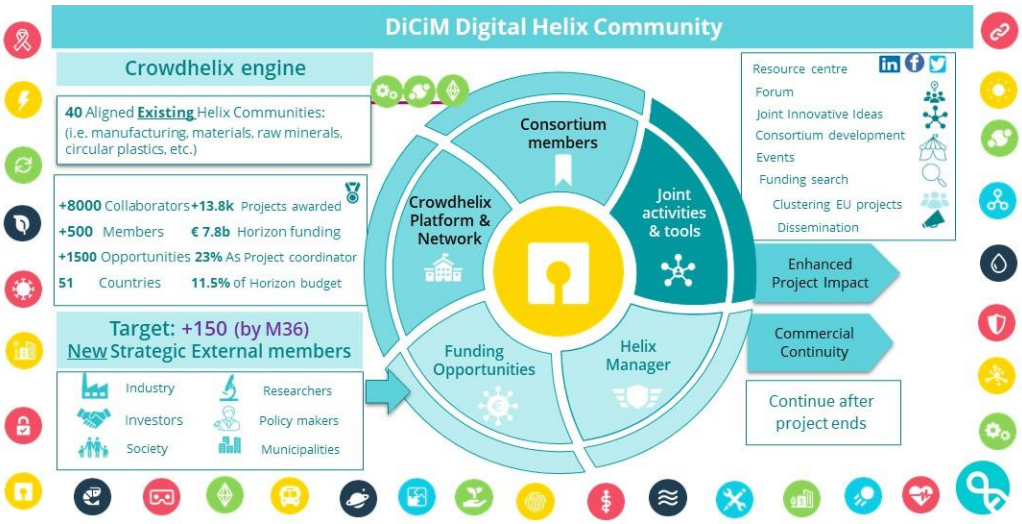

Figure 8: DiCiM Digital Helix: Virtual Community/ecosystem

→ *DiCiM Clustering plan*

DiCiM cross-projects collaboration plan (aligned to task 8.6) is divided in 3 main sections, as shown in **Figure 9**:

Collaboration board (S1): The projects with similar aims and synergies will be identified to create joint groups (1.6). To achieve this DiCiM will collaborate with relevant EU projects (included in Table 9), and relevant actors/associations across Europe/International (i.e., Gaia-X, EFFRA, Processes4Planet, EIT Manufacturing, EIT Digital, S3P-Industry, etc.). All the projects and key stakeholders will be contacted (1.1) and the Collaboration Board (CB) will be consolidated (1.2). The CB will define the Key Performance Indicators (KPIs) (1.3) and the main ways of a fluent and frequent communication (1.4). A common identity (i.e logo) will be designed (1.5).

Joint activities (S2): the CB members will promote and encourage collaborative activities (mainly with the sister projects), including regular meetings (2.1), joint participation and attendance to scientific conferences and business tradeshows (2.2), active participation in projects social media sites (2.3), development of joint scientific publications (2.4), and supporting the development of future legislation around digital tools (2.5).

Future collaboration plan (S3): a virtual workshop will be organised to promote the development of interdisciplinary consortiums, and to identify target calls aligned with their strengths. The Digital Helix will be an engine of collaborative activities.

The consortium partners will publish 10 publications by project end in scientific journals (e.g. Nature Communications) and present the results at media related congresses. We will interact with running projects DigiPrime (C-ECO as liaison) and CirThread (GOR and HGE as liaison).

Figure 9: DiCiM Clustering plan

→ *Individual dissemination plans*

Table 18: Individual dissemination plans.

Partner	Specific dissemination, exploitation and communication activities foreseen

CHX	-CHX will lead the dissemination and communication activities (development/management of the project logo, website, social media sites, press releases, newsletters, video, etc.) and WP8. -CHX, in collaboration with the consortium partners, will build the Digital helix, a virtual community/ecosystem on the Crowdhelix Platform, expecting to reach 150 new members (project partners, researchers, industry actors, policy makers, civil society, participants of other relevant projects, etc.) by M36. -CHX will organise 1 helix event. -CHX will promote clustering activities with relevant European projects and initiatives.
MU	- Write 4 peer-reviewed articles in high-quality international journals - Present the research findings 4 large international events and conferences in Europe and around the world (SME annual conferences in the US, Ellen MacArthur Foundation, world circular economy forum and annual conferences worldwide) - Conduct and publish at least 1 press release - Support peer-reviewed open-access articles written by academic and research partners - Produce 4 presentations in relevant scientific conferences - Prepare dissemination material, both digital (banners) and printed (flyers, posters) - Disseminate the research findings via the academic and industrial network of partners around the world (contact big 10 companies in Europe and countries around the world (the US and east Asia) - Contribute to the construction, preservation and updating of the webpage that will be built for the project. - Contribute to conducting and distributing weekly or monthly newsletters to subscribers on the official webpage of the project - Organise at least 8 industrial/ research workshops - Develop education materials reaching up to 250 students /year - Involve at least 2 PhD students - integrate industrial cases into the lifelong learning courses - Use social media channels (such as https://www.linkedin.com/company/76782551) and the web page of such as https://www.kth.se/en and https://www.iip.kth.se/ to disseminate the project, its activities and the research findings (create and promote success stories). MU, TUC and ULFS will use similar channels that they have access to.
KTH	
TUC	
ULFS	
IDE	- Support the peer-reviewed open-access articles written by academic and research partners. And write a peer-revied publications and white papers as confidentiality allows it. - Use corporate social media channels and company webpage to disseminate project results - Present the project results at local and international conferences - Present the project in their respective industry associations to foster the uptake of project results. -Present the results to customers and partners - Webinars
IRIS	
SIG	
C-ECO	
GOR	
ARCE	
LEX	

Table 19: Joint communication, dissemination, and training activities

Target Audiences	Target objectives	Tools & channels
• Main customers: key actors in the European manufacturing sector. Engineers, technicians, and operators on the factory floor. • Other cross-cutting HEU projects, international and national projects. • EFFRA, Processes4Planet, industry multipliers, Sectorial Associations and E**C** initiatives such as the European Platform on Life Cycle Assessment . • Standardization bodies. • General Public and Mass Media.	• Project awareness generation. • Dissemination of novel digital technologies and materials. • Collaboration with related projects such as Waste4Think, Arrowhead Tools, AI4DI, DENiM, in addition to those from Table 9 • Create awareness of R&D activities for new materials and technologies. • Communication to policymakers to increase the influence of the project and its results and in future policies. • Communication to the general public about efforts being made towards the recognition of refurbished and remanufactured products. • Effective knowledge sharing among	• Project corporate identity tools (logo, slogan) along with project website, posters, pitch decks, press releases. • Conferences, events, seminars and workshops: attending at least 2 events and organising 4 seminars/workshops. • 22 articles in Open access journals, mass media and magazines. • Social media: account on Twitter and LinkedIn mainly, we will also consider presence on other social media like Facebook and YouTube. • Visibility on third party websites (Research Gate, SlideShare, etc) • EU dissemination sources such as the

• Research and academic stakeholders. • Early-stage researchers. • PPPs, EIPs and EITs (i.e. EIT Manufacturing, EIT Digital, S3P-Industry).	scientific communities. • Attract young talent to science. • Motivate early-stage researchers to remain in the EU and investigate. • Collaboration Increase private participation in R&D activities and attract valuable stockholders.	EU Horizon Magazine, Innoradar or Innoboost. • Features in general newsletters and e-bulletins for society awareness of renewable energy storage technologies (appear in at least 4 relevant bulletins) • Digital Helix community/ecosystem

Table 20: Communication and dissemination KPIs.

Track	M12	M24	M36	M48	KPI (cumulative)
Social Media Campaigns	Twitter LinkedIn Facebook	Twitter LinkedIn Facebook	Twitter LinkedIn Facebook	Twitter LinkedIn Facebook	Twitter followers = 400 LinkedIn group members = 200 Facebook followers = 400
Podcast				6 episodes	Soundcloud reproductions = 1200
Project Video	Video1	-	-	-	Total YouTube Views = 2,000
Press releases	Press release 1	Press release 2	Press release 3	Press release 4	Total PR coverage (incl. online articles) = 500
Scientific Publications	2 submissions	4 submissions	4 submissions	6 submissions	Total publications = 16
Articles	5 articles	10 articles	10 articles	10 articles	Total articles = 35
Industrial Workshops	-	1 workshop	2 workshops	4 workshops	Total attendees = 500
Seminars		5 seminars		5 seminars	Total attendees = 500
Digital helix	50 members	50 members	50 members	Free join	Minimum members=150

2.2.3. IPR strategy related to the result type.

We are strongly committed to a sound IPR strategy that is advantageous for all parties and easy to understand. Dissemination and use of knowledge generated in the project are governed by the Grant Agreement and the Consortium Agreement terms. To make sure that these terms are followed, to avoid disputes and facilitate business planning, the Steering Committee will maintain an IPR register throughout the lifetime of the project. This document will list all items of knowledge relating to the work of the project (both background know-how and results developed in the project) and make explicit for each item its owner, nature, status and dissemination and protection measures. The directory will be regularly updated and distributed to all partners. It will form a key tool to enable knowledge management. An initial version of the IPR register will be created at the start of the project. However, at the stage of producing the proposal, the consortium has already considered what kind of strategy should be followed concerning IPR issues for the project's main results and reached a preliminary agreement on this. The basic principle on which we are agreed is that research and development results must be available to a large audience to facilitate the broad adoption of project results while having options in place for rewarding those that invested. "IPR" means that the contributing partner will own the IPR of the result it has contributed to. "Use rights" mean that any third party (licensee) is allowed to request a license to use the IPR generated by another partner (licensor) against a financial compensation to be agreed between the licensor and the licensee.

Table 21: IPR status of exploitation results.

Initial agreement on IP & use rights	Partner Owner	Protection	3rd party access
Data models	MU	Open Access	
Data repository used for ML training	LEX, GOR, SIG, ARCE, C-ECO	IPR	NO ACCESS
DiCiM open access digital platforms	SIG	IPR	Use rights
Extend ReCiPSS spare parts data management platform'	C-ECO	IPR	Use rights

ML algorithms	IRIS, ULFS	IPR	Use rights
AR solutions	TUC, IDE	IPR	Use rights
Image processing solution	IRIS	IPR	Use rights

2.3. Summary

Table 22: Summary of the impact section

SPECIFIC NEEDS	EXPECTED RESULTS	D&E&C MEASURES
▪ Value recovery is a critical step towards circularity that must be integrated with the value use phase with digital tools to extract its full power. ▪ Need for common ground across the manufacturing sector through synergies and standards.	▪ 4 large-scale demonstrators in relevant industry sectors, directly applicable to 67% of the European Industry. ▪ Creation of a community of early adopters that welcomes and enhances project results.	▪ Scientific articles & posters (D) ▪ Technical conferences and workshops (D) ▪ Dissemination to the broader public (D) ▪ Events with related projects (C) ▪ Public events (C) ▪ IPR protection of exploitable results (E) ▪ Reach out to industry players (E)
TARGET GROUPS	**OUTCOMES**	**IMPACTS**
▪ Researchers ▪ Policy makers ▪ Public society ▪ Industrial players in the manufacturing sector. Targeting specifically OEMS in the automotive parts, electronic goods, and white goods industries first.	▪ Increased awareness for industrial players of the gains of value recovery activities. ▪ Better public understanding and acceptance of remanufactured and refurbished products. ▪ Enhanced awareness about the importance of standards for academy stakeholders.	▪ Increased awareness of standards for the academy (scientific) ▪ Important gains for European industry as KPIs are included in **Table 3** to **Table 6** (economic) ▪ Specific materials for executives, middle managers, engineers, and operators and society to reach five million people (societal)

3. Quality and efficiency of the implementation

3.1. Work plan and resources

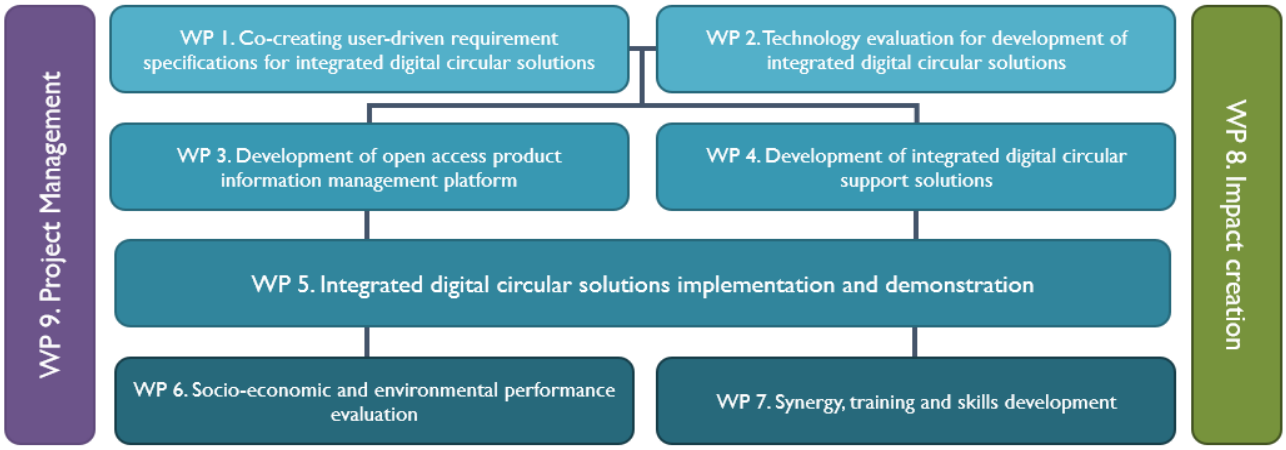

Figure 10: Pert Chart

DiCiM project will last 48 months and will be divided in 9 Work Packages (WPs). The Pert chart (**Figure 10**) provides an overview of the DiCiM work plan structure.

3.1.1. List of work packages

See Annex 1 (part A) of the grant agreement

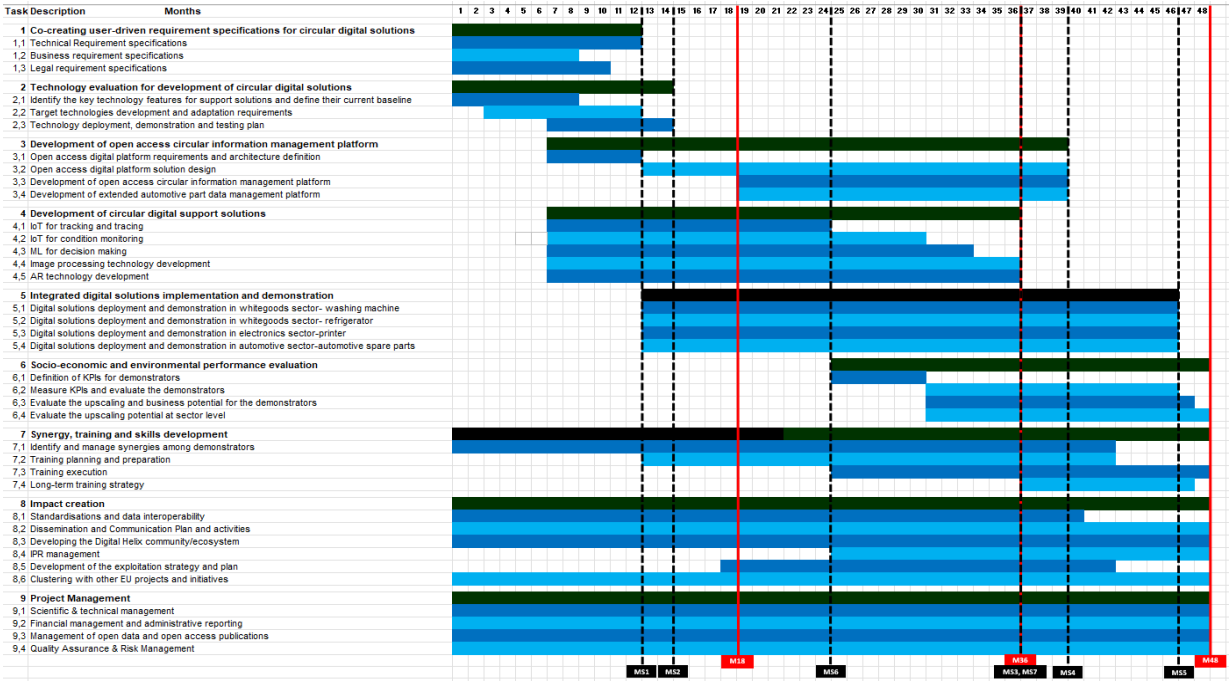

Figure 11: Gantt chart of the project

→ ***Work package description***

See Annex I (part A) of the grant agreement

3.1.2. List of Deliverables

See Annex I (part A) of the grant agreement

3.1.3. List of milestones

See Annex I (part A) of the grant agreement

3.1.4. Critical risks for implementation

See Annex I (part A) of the grant agreement

Resources

→ ***Summary of staff effort***

See Annex I (part A) of the grant agreement

→ ***'Subcontracting costs' items and 'Other costs categories' items***

No subcontracting costs envisaged for the project, nor other costs such as internally invoiced goods and services

→ ***'Purchase costs' items (travel and subsistence, equipment, and other goods, works and services)***

1/ MU	Cost	Justification
Travel	€42,000	24 person-trips over 4 years; Average trip = €1,750
Other	€38,000	open access (10k), training materials including services of graphic (10k), research data collection (raw data) - 10k, conference fees (3k), audit (5k)
Total	**€80,000**	

2/ KTH	Cost	Justification
Travel	€42,000	24 person-trips over 4 years; Average trip = €1,750
Other	€25,000	open access (12 k), Software licence (5 K) conference fees (3 k), audit (5 k)
Total	**€67,000**	

3/ TUC	Cost	Justification
Travel	€35,000	20 person-trips over 4 years; Average trip = €1,750
Other	€35,000	open access (5.5k), conference fees (3k), audit (6k); AR-tracking-software licences(7.5k); AR-Devices, associated computers, and interaction devices (10k), consumables (3k)
Total	**€70,000**	

4/ ULFS	Cost	Justification
Travel	€28,000	16 person-trips over 4 years; Average trip = €1,750
Other	€8,500	open access (5.5k), conference fees (3k)
Total	**€36,500**	

5/ IDENER	Cost	Justification
Travel	€28,000	16 person-trips over 4 years; Average trip = €1,750
Total	**€28,000**	

6/ IRIS	Cost	Justification
Travel	€28,000	16 person-trips over 4 years; Average trip = €1,750
Other	€30,000	Software licenses, server, pc, electronics and other consumables for model development and the monitoring system testing (25k), open access publication and conference fee (5k€);
Total	**€58,000**	

7/ SIG	Cost	Justification
Travel	€35,000	20 person-trips over 4 years; Average trip = €1,750
Other	€25,000	audit (5k), 20k IT-infrastructure/cloud-hosting
Total	**€60,000**	

8/ C-ECO	Cost	Justification
Travel	€28,000	16 person-trips over 4 years; Average trip = €1,750
Other	€35,000	audit (5k), 25k IT-infrastructure/cloud-hosting, 5k AR-devices (data-goggles)
Total	**€63,000**	

9/ GOR	Cost	Justification
Travel	€35,000	20 person-trips over 4 years; Average trip = €1,750
Other	€50,000	sensors & new functionalities development of washing machine prototype electronics (50k)
Total	**€85,000**	

10/ ARCE	Cost	Justification
Travel	€35,000	20 person-trips over 4 years; Average trip = €1,750
Equipment	€35,000	The total depreciation costs (%50) of robot (15k), data acquisition system (3k), data server (5k), camera (2k), thermal camera (2k), electrical test kits (3k), sensors for odor measurement (5k)
Other	€6,000	electronic consumables (3k), conveyor prototype materials (3k)
Total	**€76,000**	

11/ LEX	Cost	Justification
Travel	€35,000	20 person-trips over 4 years; Average trip = €1,750
Total	**€35,000**	

12/ CHX	Cost	Justification
Travel	€35,000	20 person-trips over 4 years; Average trip = €1,750
Other	€36,038	Leaflets and labelled folders (€10k), 1 professional video (8,610€), 1 Helix event: €11k (event room, food, drink, etc). Website: €6428 (domain purchase and maintenance)
Total	**€71,038**	

Additional justification:

The consortium has budgeted for 232 person-trips which are detailed below.

The DiCiM project is planning to have 9 General Assemblies (GAs) over 4 years. In each GA, on average 1.5 persons/partner is estimated to participate which adds up to 162 person-trips. 20 person-trips are estimated for the review meetings.

Besides, we are expecting 30 bilateral technical meetings among partners adding up to 30 person-trips.

We are also expecting 20 person-trips for dissemination activities including scientific conferences.

All the partners confirmed that all costs related to equipment include only the depreciation costs for the mentioned equipment.

3.2. Capacity of participants and consortium as a whole

The consortium consists of 12 partners and 2 advisory dodies (**Table 23**) – gathering a unique mix of leading industry expertise and knowledge to develop and demonstrate integrated digital solutions to improve the efficiency and responsiveness of recovery value activities and for a successful implementation of the **Circular Manufacturing System**.

Table 23: DiCiM consortium and supporting stakeholders

The project will be coordinated by MU, that was already involved in the ReCiPSS flagship project, and has a strong track record in open science practices: following the principle of open access, all research results of scientists at MU are to be made freely accessible throughout the world and thus be rendered as visible as possible. MU supports all members of the university by means of general financial and structural conditions that have been created to achieve this objective. The project manager at MU will be Mr. David Póč, the Chief Strategy Officer at the Faculty and the Strategic Project Director at the University (rectorate). He has a background in Public Administration, History and Philosophy and long experience as the Head of the Project management team at the Faculty of Administration and Economics, responsible for the successful realization of all projects both at the national and international levels.

Gender balance is aimed at all levels of project management. Both genders are represented at the task leader and group level, **with women leading 4 of the 9 work packages.**

Figure 12: Commitment letters from ERC (left) and APRA (right)

APRA and ERC, the Automotive Parts Remanufacturers Association and the European Reman Council have an advisory role in the project. Both have committed to support the project as members of the Project Advisory Board (**Figure 12**), and participate in the joint dissemination and communication actions, as specified in **Table 19**.

The consortium includes the key disciplines described in Table 10. With this composition, the consortium will be able to:

• Develop refined, improved and/or new ML, DLT based and AR models and algorithms to unlock big data potential for value recovery in Europe (MU, KTH, TUC, ULFS, IDE, IRIS, SIG, C-ECO)

• Aid the SMEs to develop and demonstrate of the potential of value recovery and value reuse (KTH, IRIS, SIG, C-ECO, GOR, ARCE, LEX, CHX, APRA, ERC)

• Increase the general public's understanding of value recovery products as enablers of green transition and circular economy (CHX, APRA, ERC)

• Influence decision-makers to evaluate the possibilities for standardisation in Europe in a socially, environmentally, and economically sustainable way (all)

4. Ethics

Administrative forms

Proposal ID **101091536**

Acronym **DiCiM**

4 - Ethics & security

Ethics Issues Table

1. Human Embryonic Stem Cellsand Human Embryos			Page
Does this activity involve Human Embryonic Stem Cells (hESCs)?	○ Yes	◉ No	
Does this activity involve the use of human embryos?	○ Yes	◉ No	
2. Humans			**Page**
Does this activity involve human participants?	◉ Yes	○ No	36-37
Are they volunteers for non medical studies (e.g. social or human sciences research)?	○ Yes	◉ No	
Are they healthy volunteers for medical studies?	Yes	◉ No	
Are they patients for medical studies?	Yes	◉ No	
Are they potentially vulnerable individuals or groups?	Yes	◉ No	
Are they children/minors?	Yes	◉ No	
Are they other persons unable to give informed consent?	Yes	◉ No	
Does this activity involve interventions (physical also including imaging technology, behavioural treatments, etc.) on the study participants?	○ Yes	◉ No	
Does this activity involve conducting a clinical study as defined by the Clinical Trial Regulation (EU 536/2014)? (using pharmaceuticals, biologicals, radiopharmaceuticals, or advanced therapy medicinal products)	○ Yes	◉ No	
3. Human Cells / Tissues (not covered by section 1)			**Page**
Does this activity involve the use of human cells or tissues?	○ Yes	◉ No	
4. Personal Data			**Page**
Does this activity involve processing of personal data?	○ Yes	◉ No	
Does this activity involve further processing of previously collected personal data (including use of preexisting datasets or sources, merging existing datasets)?	○ Yes	◉ No	
Is it planned to export personal data from the EU to non-EU countries? Specify the type of personal data and countries involved	○ Yes	◉ No	
Is it planned to import personal data from non-EU countries into the EU or from a non-EU country to another non-EU country? Specify the type of personal data and countries involved	○ Yes	◉ No	
Does this activity involve the processing of personal data related to criminal convictions or offences?	○ Yes	◉ No	
5. Animals			**Page**
Does this activity involve animals?	○ Yes	◉ No	
6. Non-EU Countries			**Page**
Will some of the activities be carried out in non-EU countries?	○ Yes	◉ No	
In case non-EU countries are involved, do the activities undertaken in these countries raise potential ethics issues?	○ Yes	◉ No	

Administrative forms

Proposal ID **101091536**

Acronym **DiCiM**

It is planned to use local resources (e.g. animal and/or human tissue samples, genetic material, live animals, human remains, materials of historical value, endangered fauna or flora samples, etc.)?	○ Yes	◉ No	
Is it planned to import any material (other than data) from non-EU countries into the EU or from a non-EU country to another non-EU country?For data imports, see section 4.	○ Yes	◉ No	
Is it planned to export any material (other than data) from the EU to non-EU countries? For data exports, see section 4.	○ Yes	◉ No	
Does this activity involve low and/or lower middle income countries, (if yes, detail the benefit-sharing actions planned in the self-assessment)	○ Yes	◉ No	
Could the situation in the country put the individuals taking part in the activity at risk?	Yes	◉ No	
7.Environment, Health and Safety			**Page**
Does this activity involve the use of substances or processes that may cause harm to the environment, to animals or plants.(during the implementation of the activity or further to the use of the results, asa possible impact)?	Yes	◉ No	
Does this activity deal with endangered fauna and/or flora / protected areas?	Yes	◉ No	
Does this activity involve the use of substances or processes that may cause harm to humans, including those performing the activity.(during the implementation of the activity or further to the use of the results, asa possible impact)?	Yes	◉ No	
8. Artificial Intelligence			**Page**
Does this activity involve the development, deployment and/or use of Artificial Intelligence? (if yes, detail in the self-assessment whether that could raise ethical concerns related to human rights and valuesand detail how this will be addressed).	Yes	◉ No	
9. Other Ethics Issues			**Page**
Are there any other ethics issues that should be taken into consideration?	○ Yes	◉ No	

I confirm that I have taken into account all ethics issues above and that, if any ethics issues apply, I will complete the ethics self-assessment as described in the guidelines How to Complete your Ethics Self-Assessment ☒

Ethics Self-Assessment

Ethical dimension of the objectives, methodology and likely impact

No personal and sensitive data will be collected without the explicit and informed consent of the human participant and every information will be later accu rately pseudonymised and/or aggregated. As regards with secondary use of data (i.e. further processing of previously collected personal data) by Consortium partners, existing datasets will originate from a public source or be authorised for use in the research (either specifically for the project or generally for any secondary use). If there is the case, The Consortium will create a recruitment environment that is not only effective but isalso ethical, in the sense that maximises benefitsand minimises the risk of harm (psychological, social, legal, economic, environmental, etc.). The methods will unlikely result in discriminatory practices or unfair treatment and complies with national and EU regulations. Participants will be recruited by letter or email including: info about how the person was identified; who is doing the research and why; what is involved and for how long; an overview of any risks or potential benefits; info about how to contact the research team if interested in participating; where to get answers to additional questions, and when using email, proper notification about the confidentiality issues associated with email communication. After having proactively shown their interest in joining the project, datasubjects will be informed about the data processing and asked for their free, explicit and informed consent. All participants will be given a detailed information sheet and an informed consent form.

Any document provided to the participants will be written in terms participants can understand and translated in the national languages if needed. Participation in the activities will always be voluntary, and participants will explicitly be informed that they can refuse to participate or withdraw their consent at any time without any consequences. The procedures that will be followed to obtain informed consent are: i) participants will be welcomed; ii) participants will be informed about the project and the specific activity in which they are involved, in an appropriate format according to their needs and the approved models of the consortium. Participants will be requested to give their explicit consent. The consent form and information sheet should be included on asingle piece of paper (both sides, if needed). We will not involve persons who have not attained the legal age for consent to participate in the research. Once the activity is finished, both the information sheet and the signed consent form will be sent by recorded mail to the Data Controller (to the email that will be identified at the beginning of the project).

Compliance with ethical principles and relevant legislations

Responsibility for protecting personal information under GDPR will extend to data processing as well as data controllers.

Each consortium member agrees to comply with all obligationsand requirements of its corresponding national data protection legislation and the Data Protection Directive 95/46/EEC and to obtain all legal documents and certifications required for compliance with such legislation. The consortium membersagree that any background, results, confidential information and/or any and all data and/or information that is provided, disclosed or otherwise made available between the consortium member during the implementation of the project and/or for any exploitation activities ("Shared Information"), may include personal data as defined by Article 2,Section (a) of the Data Protection Directive (95/46/EEC), and applicable local implementing legislation (hereinafter referred to as "Personal Data").Each consortium member who provides or otherwise make available to any other Party Shared Information containing Personal Data ("Contributor") will represent that: (i) it has the authority and/or the authorisation to disclose the Shared Information, if any, which it provides to the consortium member; (ii) where legally required and relevant, it has obtained appropriate informed consents from all the datasubjects involved, or from any applicable institution, all in compliance with applicable regulations; and (iii) there is no restriction in place that would prevent any such another party from using the Shared Information in accordance with and for the purpose of the project.

The Data Controller will be responsible for complying with all applicable data protection or similar laws regulating the processing of any Personal Data (assuch terms are defined in Directive 95/46/EC) provided by or through each Data Controller to the Data Processors.Each Data Processor will only process such Personal Data in a manner which is reasonably necessary to implement the Action and exploit the Results, and only for that purpose. Each Data Processor will follow the relevant Data Controller's reasonable processing instructions with respect to the Personal Data and the Data Processors will use their reasonable endeavours to apply the security measures set forth in the project, or as notified to the Data Processors in writing in advance. The Data Controllers are responsible for determining that these measures provide an appropriate level of protection.Either Data Processor shall not process any Personal Data unless (i) it isacting on the instructions of the Data Controllers (ii) it isacting to implement the project or exploit its results; or (iii) it is required to do so by applicable law or any court order,

in which case the relevant Data Processor shall inform the Data Controller(s) immediately in writing about such disclosure.
The Data Processors, in implementing the project or exploiting its results, may transfer data, including Personal Data, across a country border, including outside the European Economic Area (EEA), if the Data Processor reasonably considers such transfer appropriate or useful for the Data Processors' implementation of the Action and exploitation of the Results and reasonably cooperates with the Data Controllers to meet legal requirements. The Data Controllers are solely responsible for determining that any transfer by the Data Processors or Data Controllers of data, including Personal Data, across a country border complies with the applicable data protection laws including assessing, determining and fulfilling any legal prerequisites or obtaining any required authorizations. The Data Controllers agree that implementation of the project or exploitation of its results will be partially performed from abroad, and may entail that Personal Data are transferred outside the EEA. In view of this data transfer, each Data Controller will enter into aso-called Transfer Agreement (EUStandard Contractual Clausesas defined in the Commission Decision of February, 5th 2010, or any amendments thereto) with the Data Importer(s) in order to secure an adequate level of protection.Each Data Controller agrees to indemnify, defend and hold harmless the Data Importer from and against any claim by a third party arising under or relating to the Transfer Agreement.